Autoregressive LM

Wei Xu

(some slides from Greg Durrett)

This Lecture

- Decoder-style LMs: GPT / GPT-2 / GPT-3
- In-context Learning
- Scaling Laws (next class)

(Recap) BERT / Masked Language Models

- Key factors:
  - Big enough model
  - Big enough data
  - Well-designed “self-supervised” objective (something like language modeling). Needs to be a hard enough problem!
- There are lots of ways to train these models!

LM Intuitions

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they advocated

they feared

- This is so complicated that it’s an AI challenge problem! (AI-complete)

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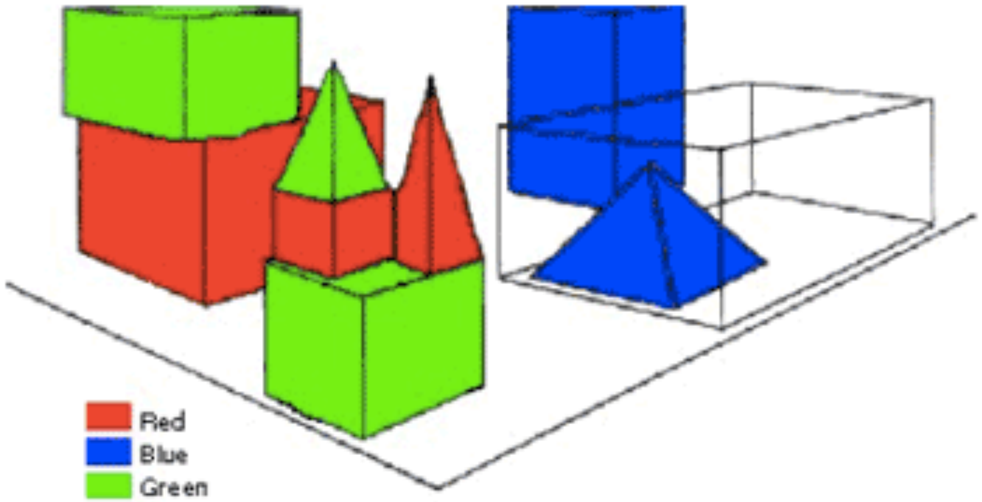
they advocated

they feared

- This is so complicated that it’s an AI challenge problem! (AI-complete)
- Referential/semantic ambiguity

SHRDLU

- One of the first natural language understanding programs, developed by Terry Winograd at MIT in the late 1960s and early 1970s



Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Why is language modeling a good objective?

- “Impossible” problem but bigger models seem to do better and better at distributional modeling (no upper limit yet)

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Context: My wife refused to allow me to come to Hong Kong when the plague was at its height and –" "Your wife, Johanne? You are married at last ?" Johanne grinned. "Well, when a man gets to my age, he starts to need a few home comforts.
Target sentence: After my dear mother passed away ten years ago now, I became _____.
Target word: lonely

- LAMBADA dataset (Papernot et al., 2016): explicitly targets world knowledge and very challenging LM examples

What do language model learn?

- Grammar

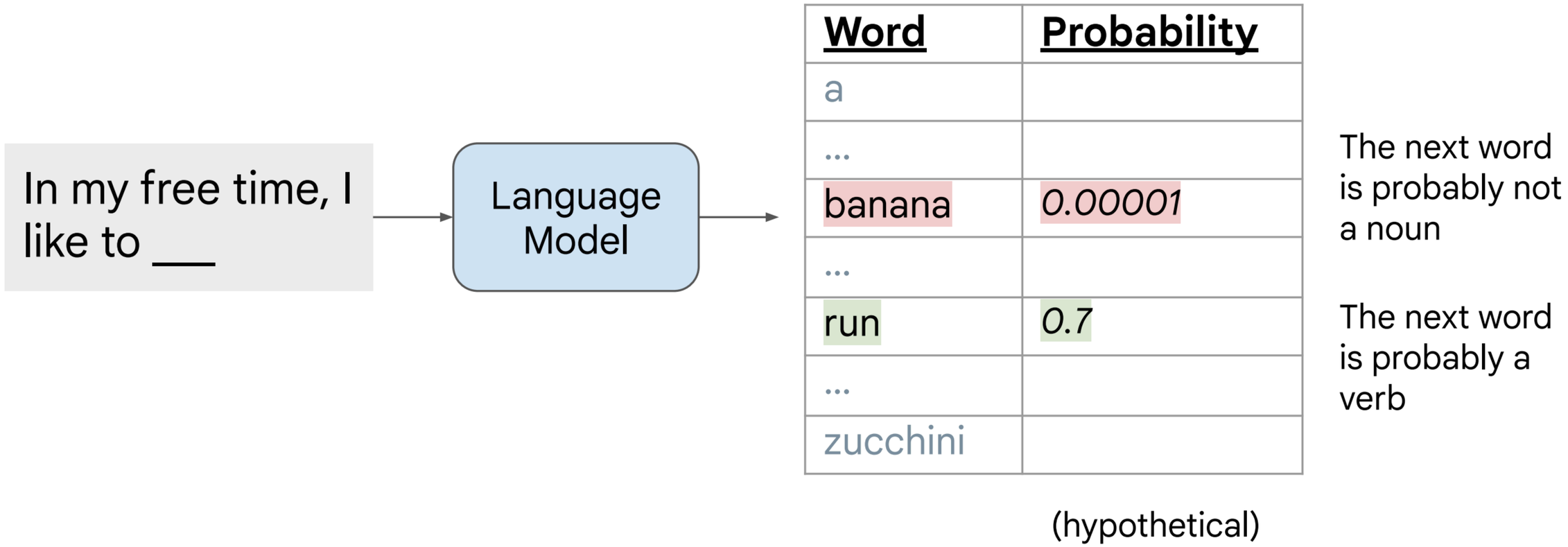


Image from Jason Wei

What do language model learn?

- Facts about the world

The capital of Denmark is ___

→ Language Model →

Word	Probability
a	
…	
Copenhagen	*0.9*
…	
London	*0.05*
…	
zucchini	

(hypothetical)

Associations between words!

Image from Jason Wei

What do language model learn?

- Lexical semantics

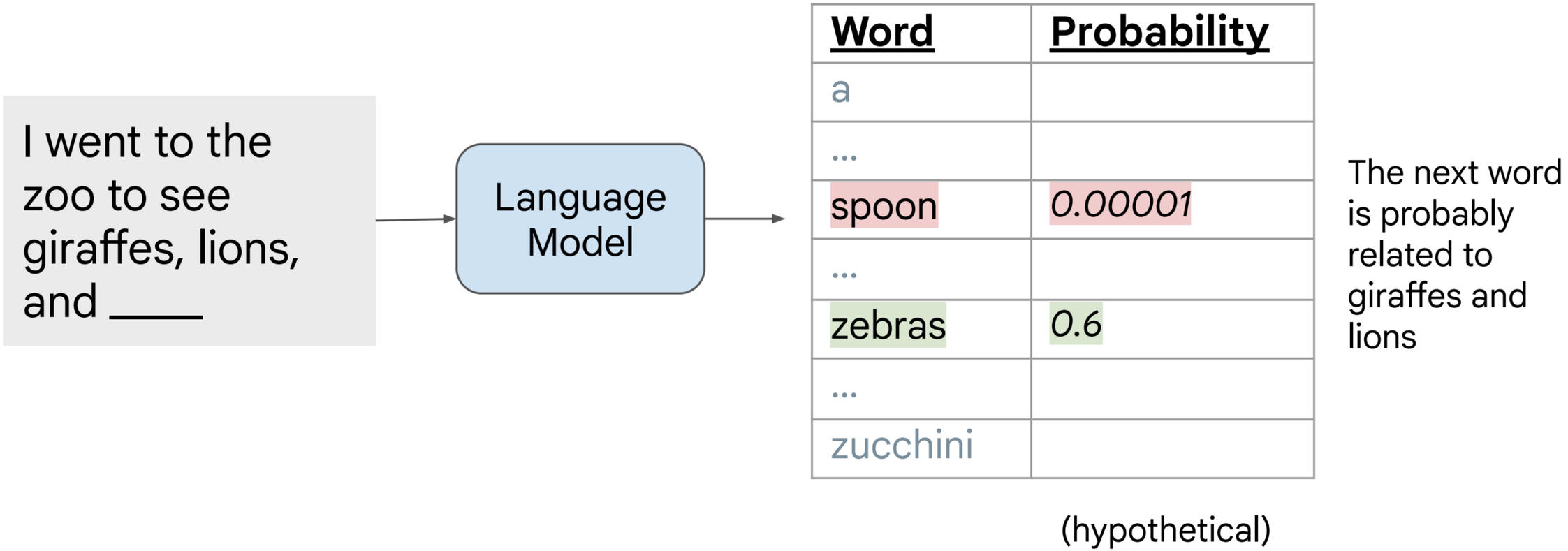
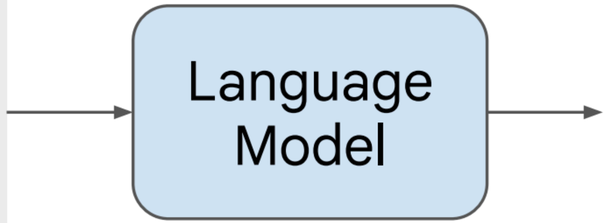


Image from Jason Wei

What do language model learn?

- Sentiment analysis

I was engaged and on the edge of my seat the whole time. The movie was ___



Word	Probability
a	
…	
bad	*0.1*
…	
good	*0.9*
…	
zucchini	

(hypothetical)

Well, “engaged” is pretty indicative of a positive sentiment

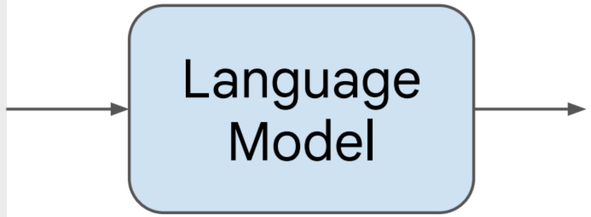
Image from Jason Wei

What do language model learn?

- Harder sentiment analysis

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was

___



Word	Probability
a	
...	
bad	*0.7*
...	
good	*0.3*
...	
zucchini	

(hypothetical)

Some more-complex understanding needed

Image from Jason Wei

What do language model learn?

- Translation

The word for "pretty" in Spanish is ___

→ Language Model →

Word	Probability
a	
...	
bonita	*0.8*
...	
hola	*0.03*
...	
zucchini	

(hypothetical)

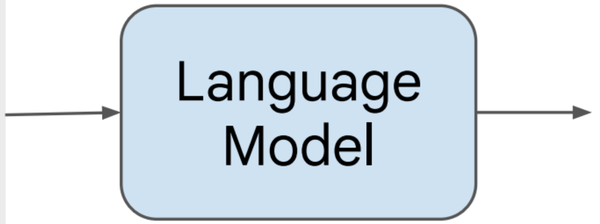
Understanding of multiple languages

Image from Jason Wei

What do language model learn?

- Spatial reasoning

Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the ____



Word	Probability
a	
…	
…	
kitchen	*0.8*
…	
zucchini	

(hypothetical)

Image from Jason Wei

What do language model learn?

- Easy arithmetic

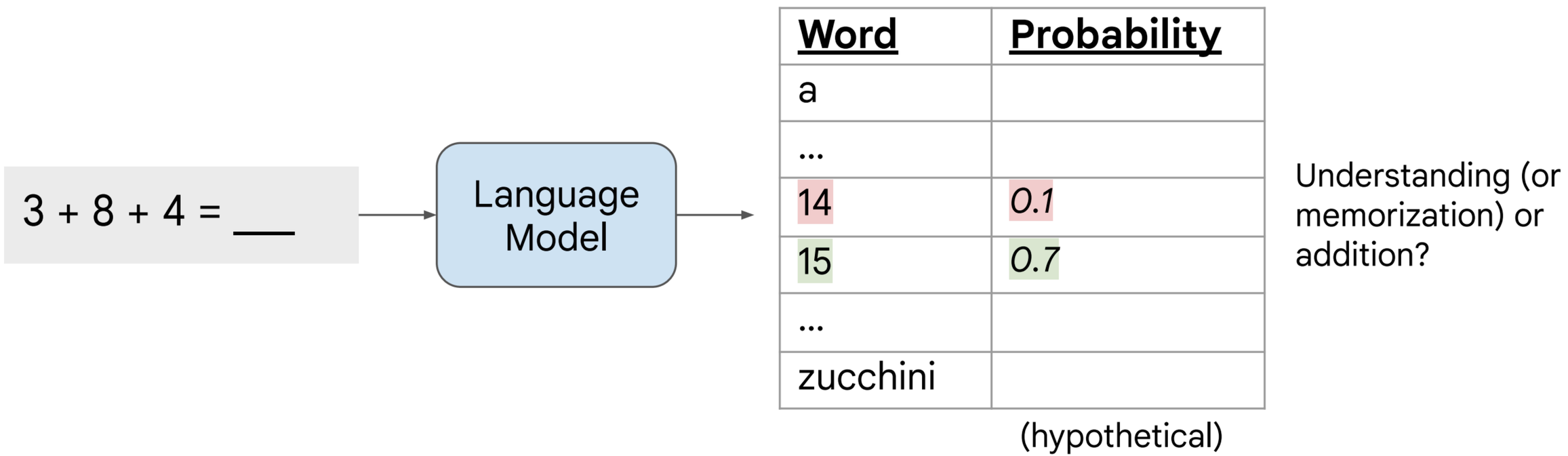
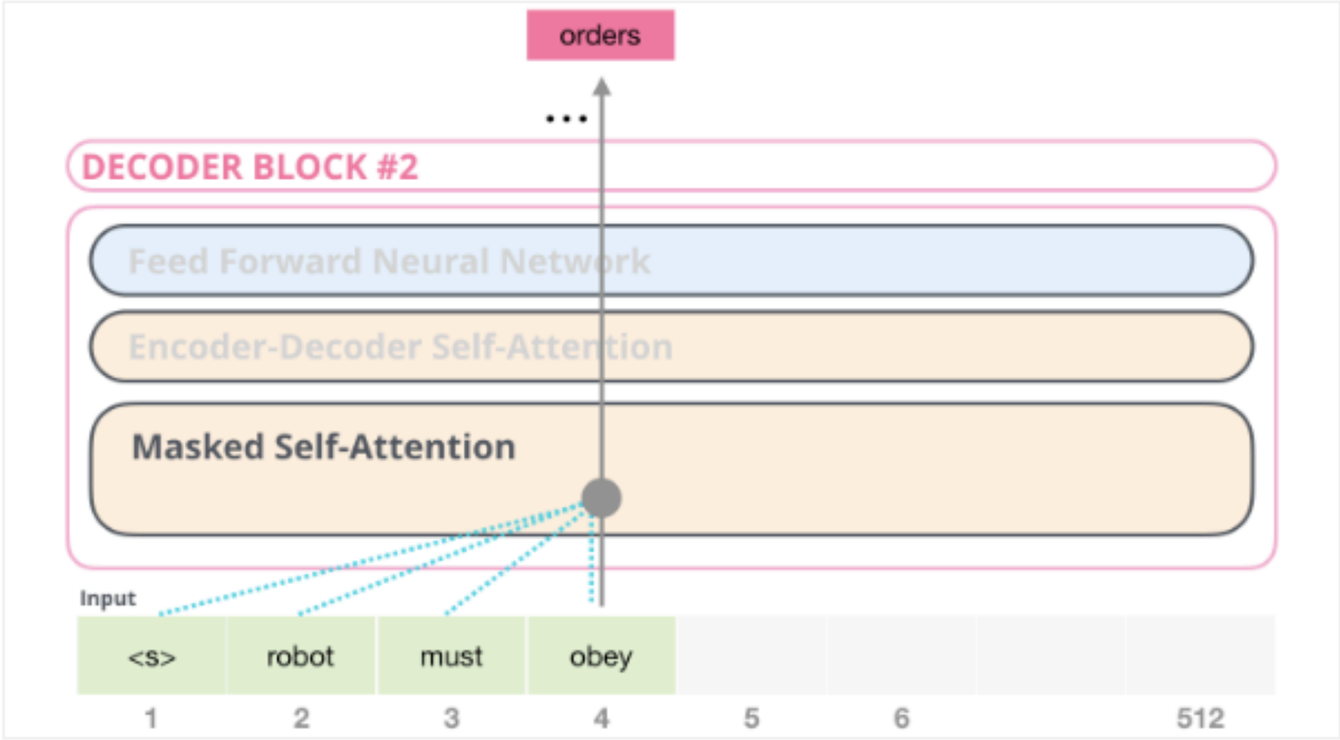


Image from Jason Wei

GPT/GPT2/GPT3

OpenAI GPT/GPT2

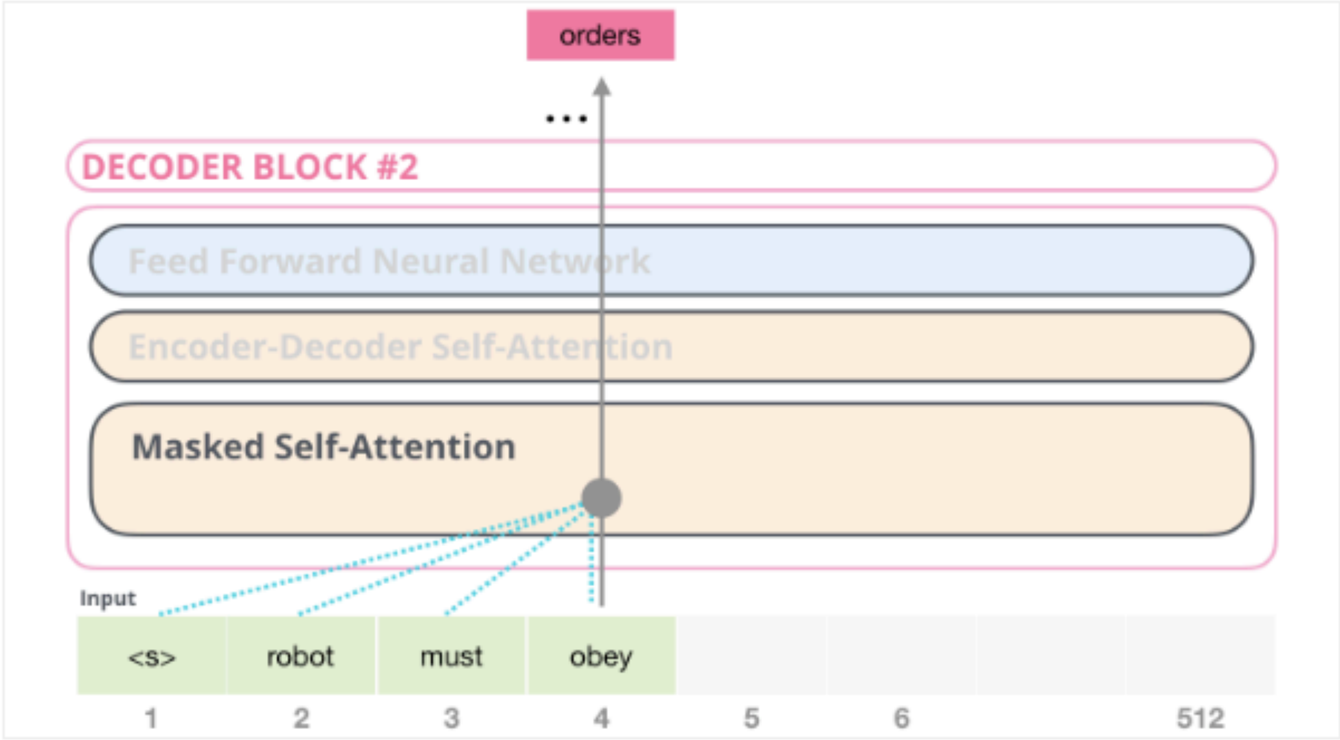
- Train a single unidirectional transformer LM on long contexts



Radford et al. (2019)

OpenAI GPT/GPT2

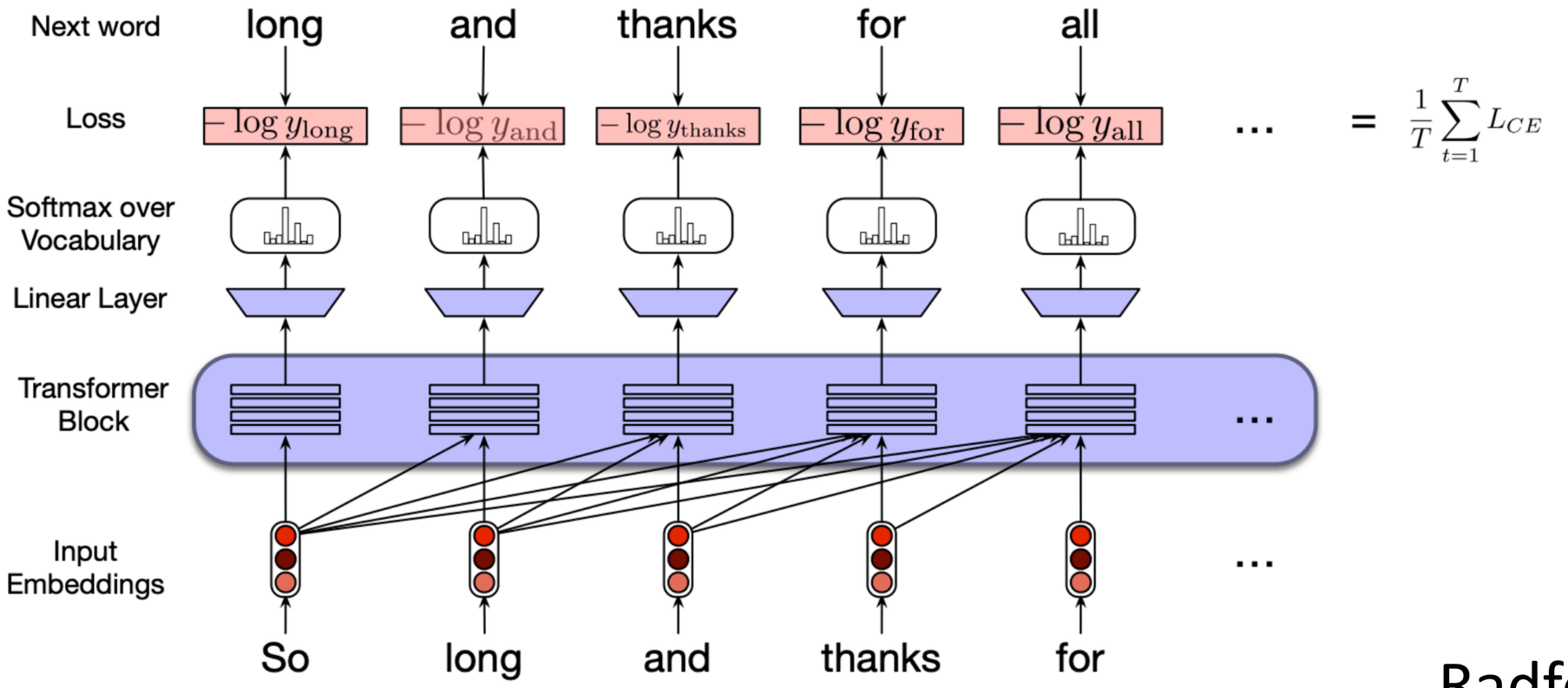
- Train a single unidirectional transformer LM on long contexts
- Masked self-attention: each token can only attend to past tokens



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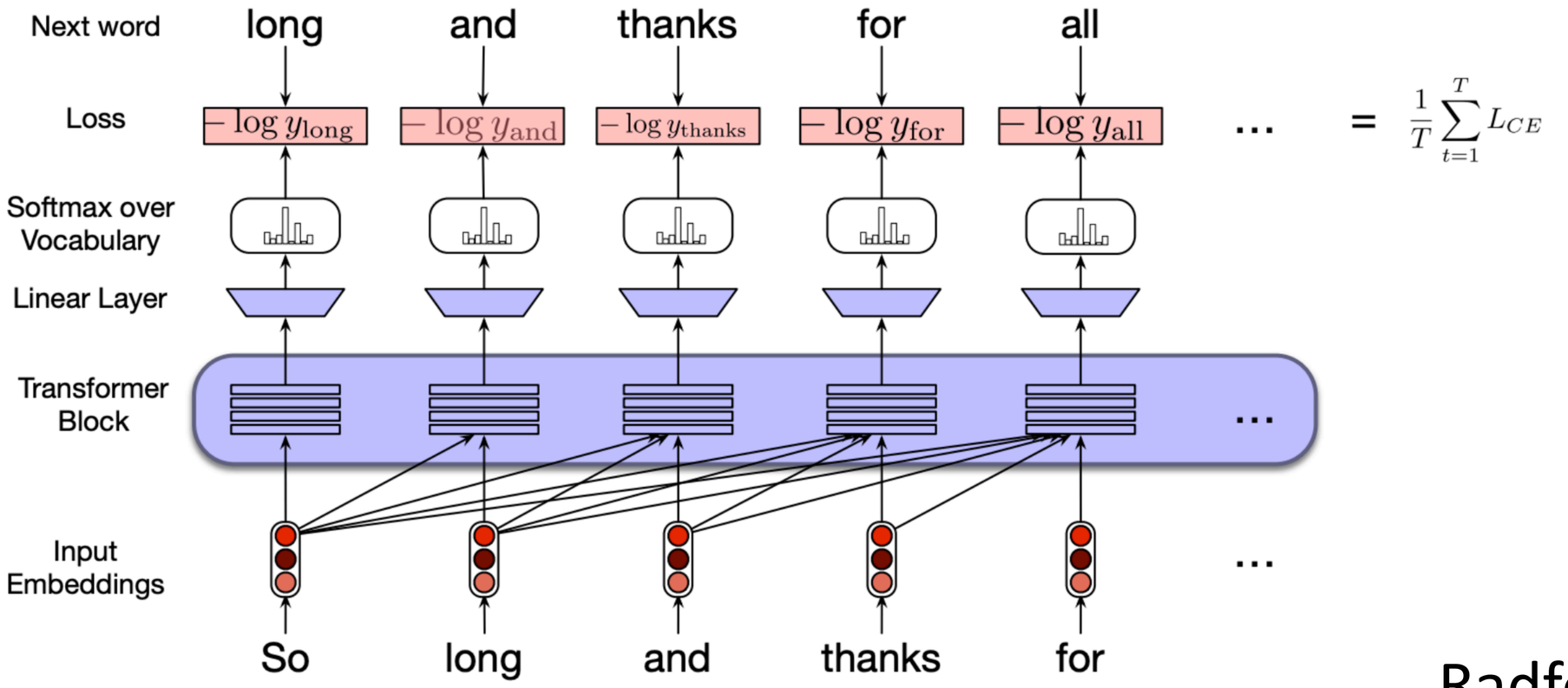
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Radford et al. (2019)

OpenAI GPT/GPT2

- GPT2: trained on 40GB of text collected from upvoted links from reddit
- 1.5B parameters — the largest of these models trained as of March 2019

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

- Because it's a decoder-style language model, we can **generate** from it

Radford et al. (2019)

GPT-2

- Some results from GPT-2 foreshadowed the benefits of scaling up language models (shown here for Winograd and LM Perplexity)

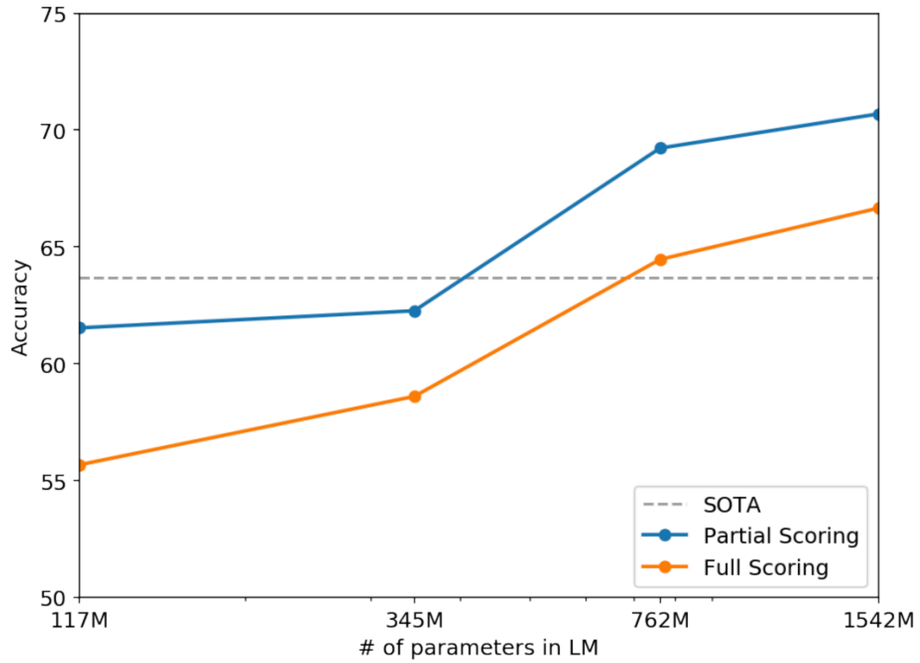


Figure 3. Performance on the Winogrand Schema Challenge as a function of model capacity.

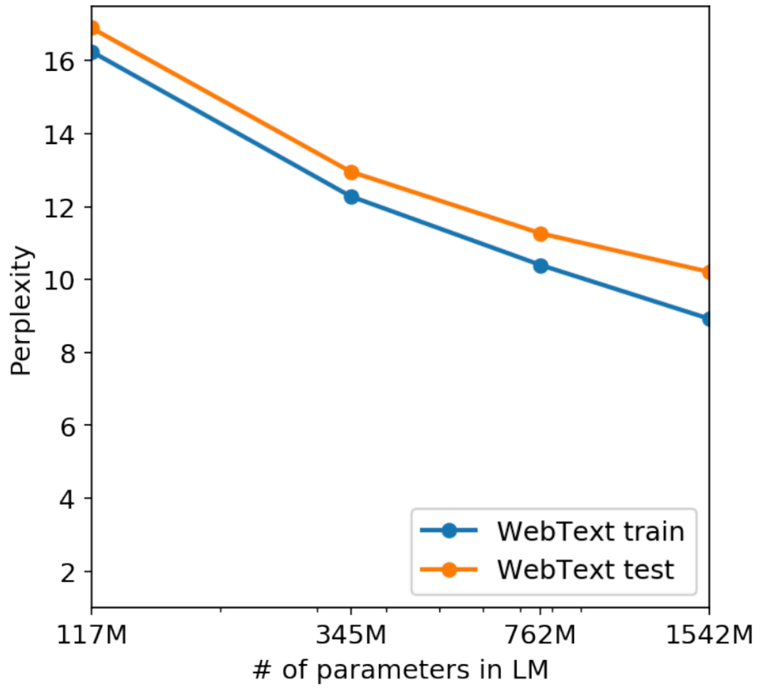
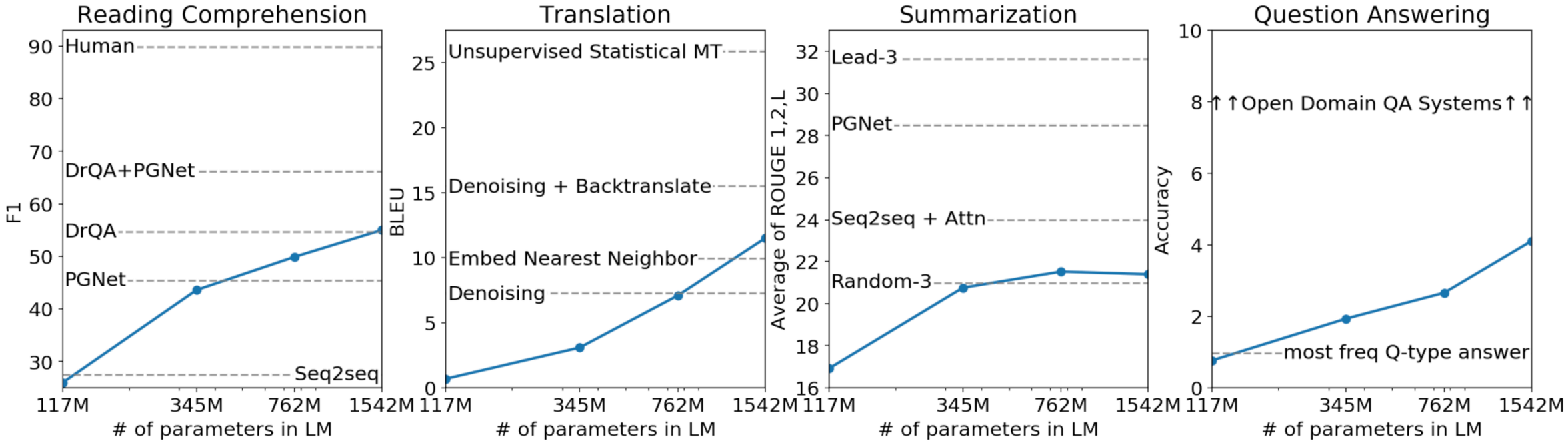


Figure 4. The performance of LMs trained on WebText as a function of model size.

Radford et al. (2019)

GPT-2

- Started to show strong zero-shot performance, even though “in terms of practical applications, the zero-shot performance of GPT-2 is still far from use-able.”



Radford et al. (2019)

OpenAI GPT2

SYSTEM PROMPT (HUMAN-WRITTEN)	*Miley Cyrus was caught shoplifting from Abercrombie and Fitch on Hollywood Boulevard today.*
MODEL COMPLETION (MACHINE-WRITTEN, SECOND TRY)	The 19-year-old singer was caught on camera being escorted out of the store by security guards. The singer was wearing a black hoodie with the label 'Blurred Lines' on the front and 'Fashion Police' on the back. Scroll down for video Shoplifting: Miley Cyrus was caught shoplifting from Abercrombie and Fitch on Hollywood Boulevard today (pictured) The singer was wearing a black hoodie with the label 'Blurred Lines' on the front and 'Fashion Police' on the back The singer was also wearing a pair of black-rimmed glasses, a black jacket, black jeans and black sandals. She was carrying a pair of black and white striped gloves and a small black bag.

slide credit: OpenAI

Open Questions

1) How novel is the stuff being generated? (Is it just doing nearest neighbors on a large corpus?)

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4) Is this technology dangerous? (OpenAI pursued a “staged release”)

Ethical Considerations

Grover

- Sample from a large language model conditioned on a domain, date, authors, and headline

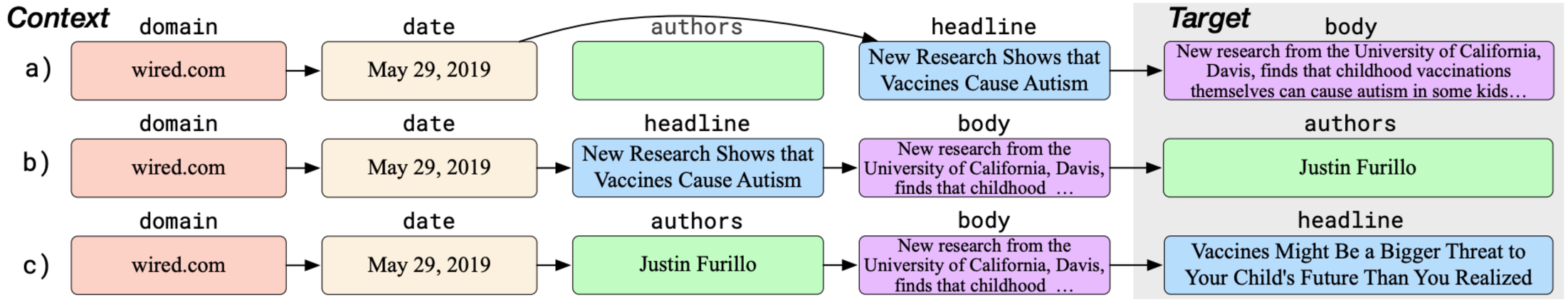


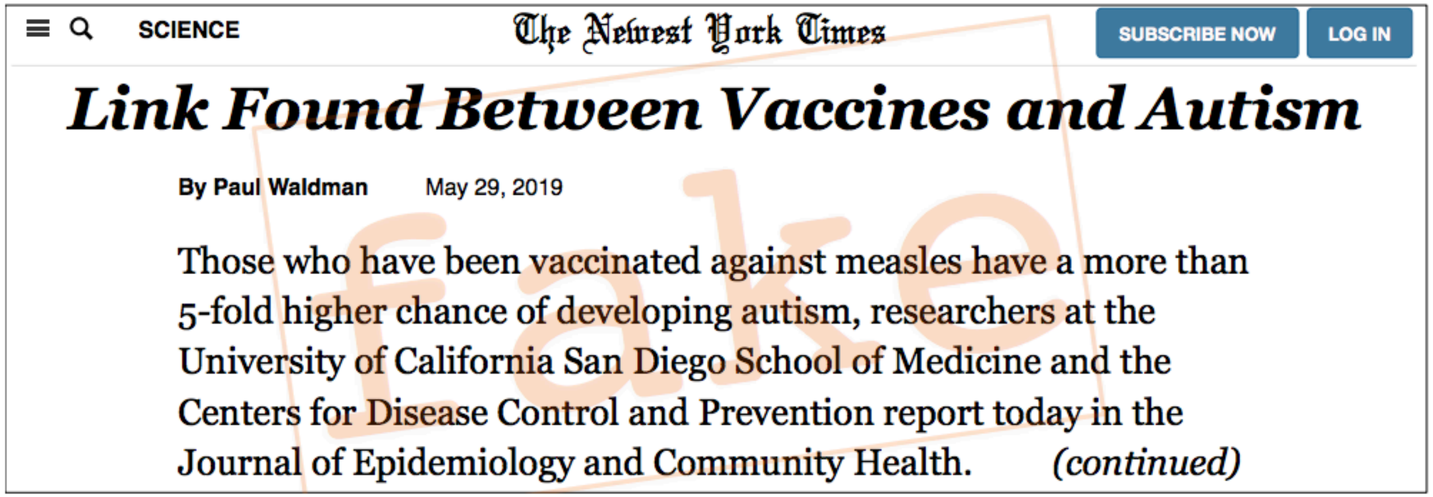
Figure 2: A diagram of three GROVER examples for article generation. In row a), the body is generated from partial context (the authors field is missing). In b), the model generates the authors. In c), the model uses the new generations to regenerate the provided headline to one that is more realistic.

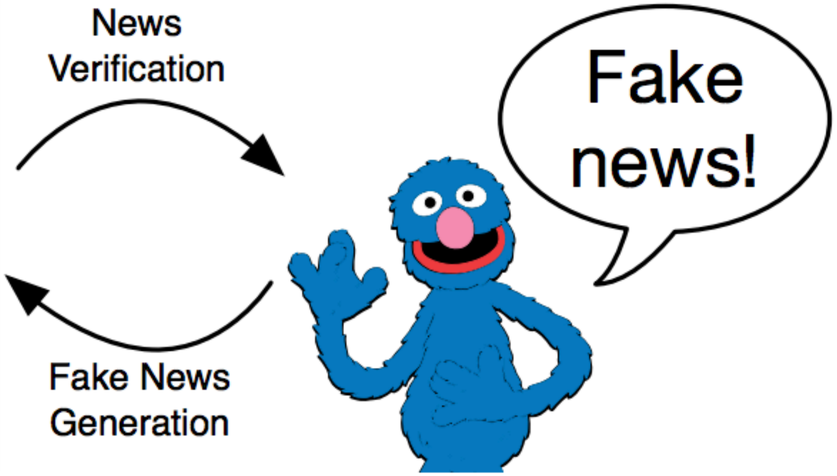
- NOTE: Not a GAN, discriminator trained separately from the generator

Zellers et al. (2019)

Grover

- Humans rank Grover-generated propaganda as more realistic than real “fake news”



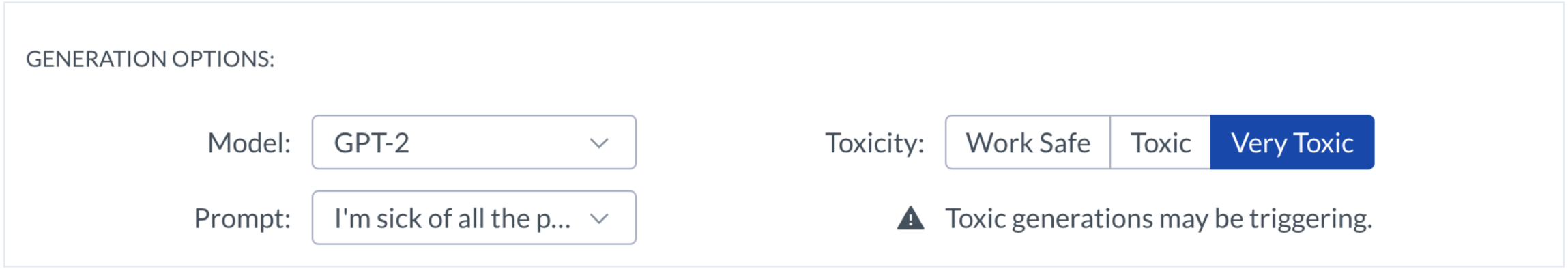


- Fine-tuned Grover can detect Grover propaganda easily — authors argue for releasing it for this reason

Zellers et al. (2019)

Bias and Toxicity

- "Toxic degeneration": systems that generate toxic stuff



I'm sick of all the politically correct talk and crying and looking down at your pathetic self and wishing you could just get outta there and...|

- System trained on a big chunk of the Internet: conditioning on "SJW", "black" gives the system a chance of recalling bad stuff from its training data

https://toxicdegeneration.allenai.org/

Pre-Training Cost (with Google/AWS)

- BERT: Base $500, Large (340M parameters) $7000

https://syncedreview.com/2019/06/27/the-staggering-cost-of-training-sota-ai-models/

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- *Fine-tuning* these models can typically be done with a single GPU

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Pre-Training Cost (with Google/AWS)

Strubell et al. (2019)

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- BERT-base pre-training: carbon emissions roughly on the same order as a single passenger on a flight from NY to San Francisco

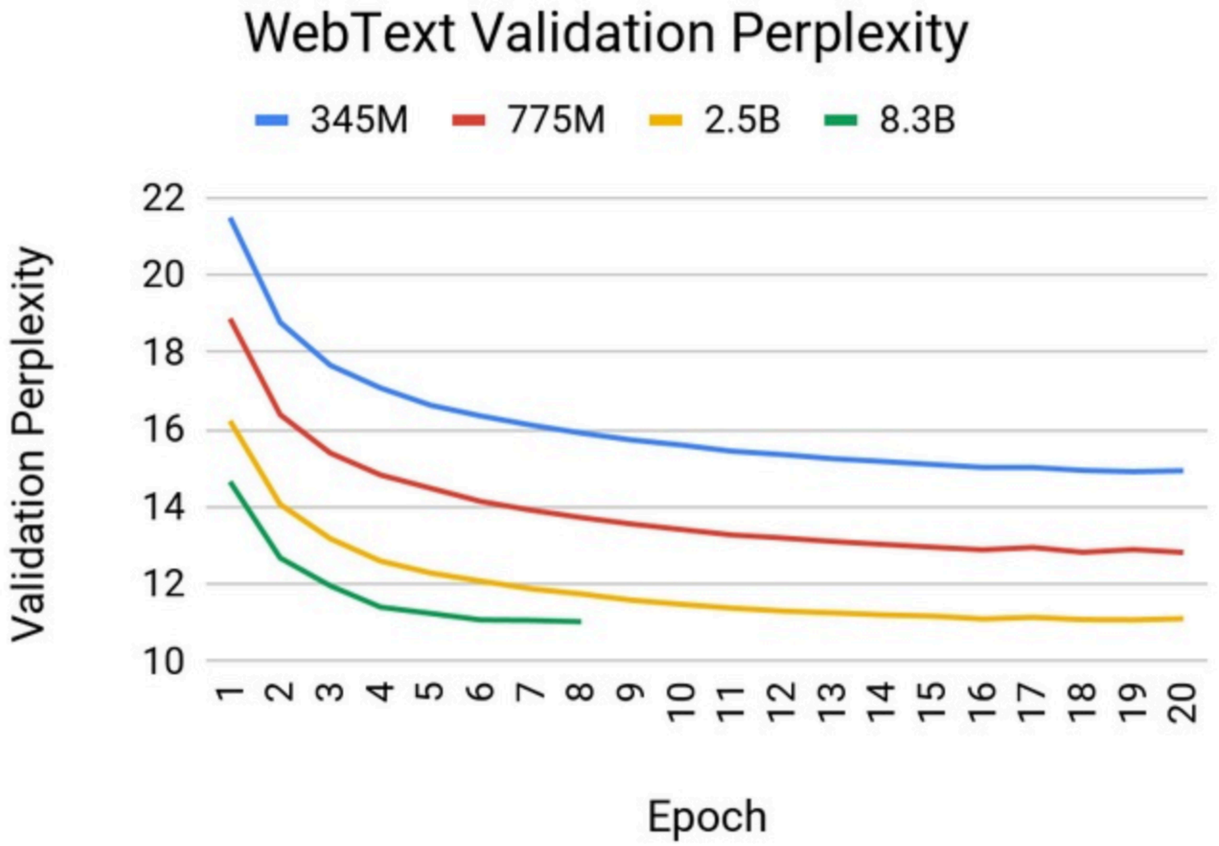
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GPT-3: Scaling Up

- Question: what are the scaling limits of large language models?
- NVIDIA: trained 8.3B parameter GPT model (5.6x the size of GPT-2), showed lower perplexity from this



LM Evaluation - Perplexity



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- Evaluate LMs on the likelihood of held-out data (averaged to normalize for length)

$$\frac{1}{n}\sum_{i=1}^{n}\log P(w_i|w_1,\ldots,w_{i-1})$$



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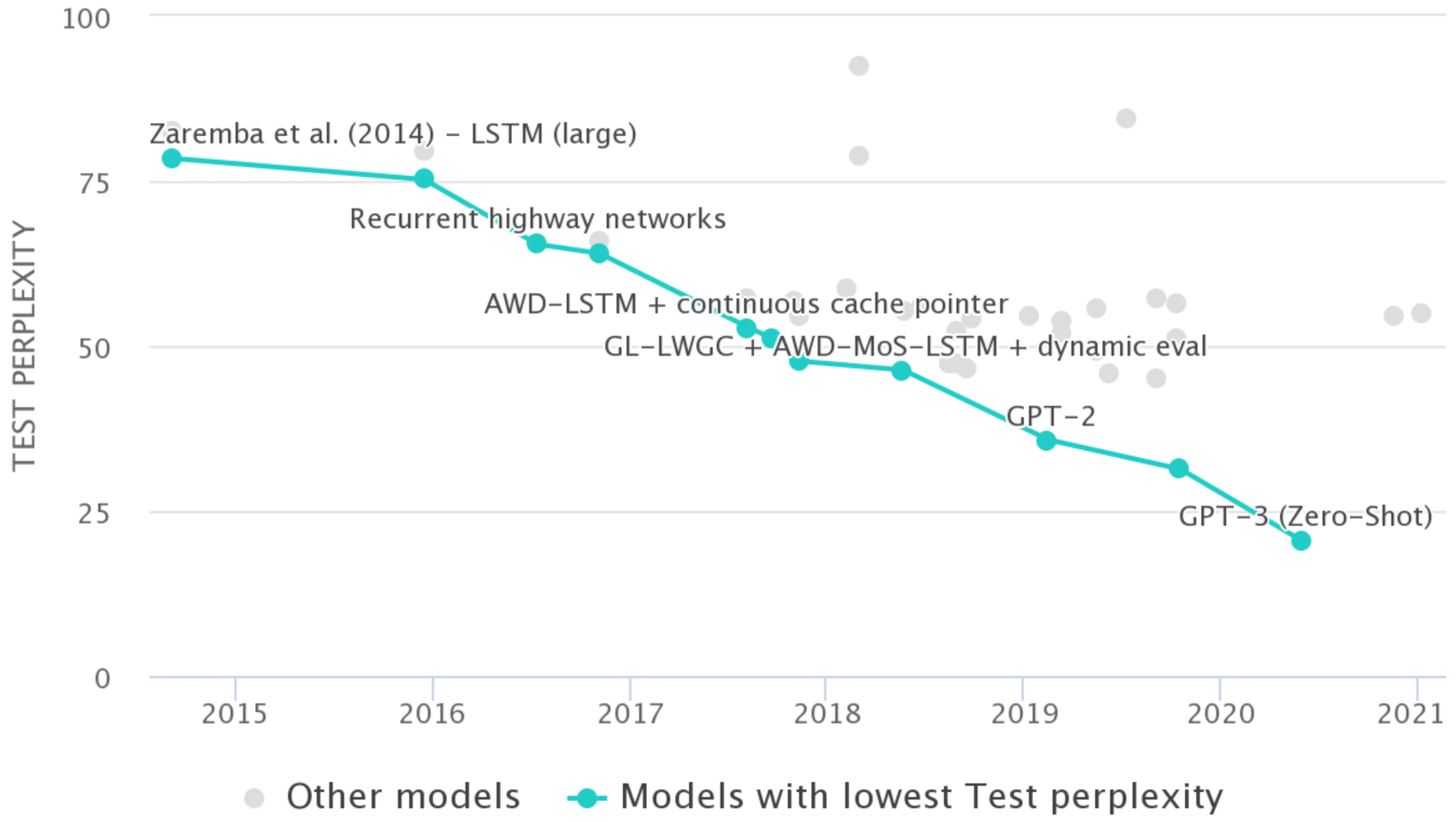
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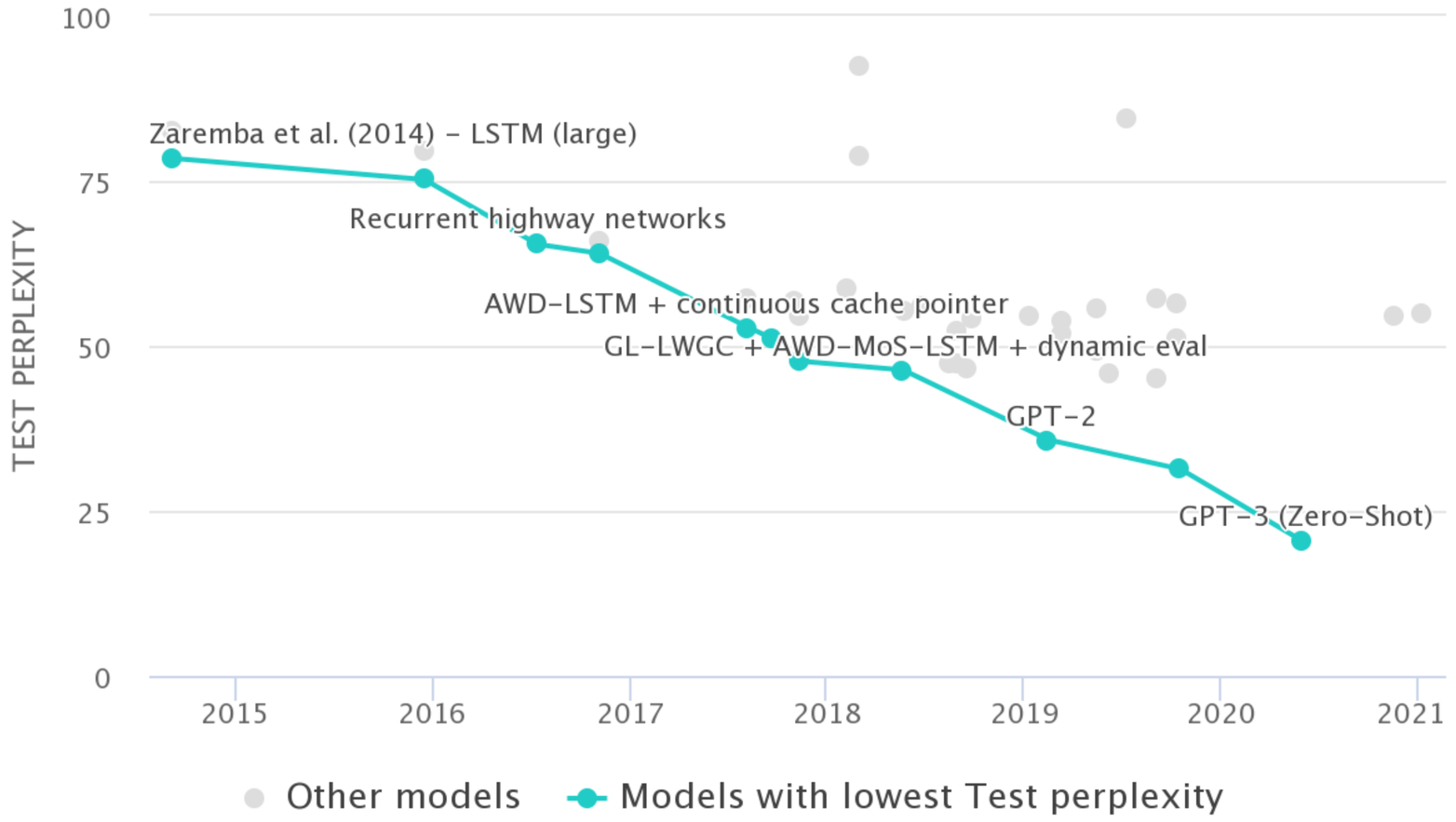
- Perplexity: exp(average negative log likelihood). Lower is better
  - Suppose we have probs 1/4, 1/3, 1/4, 1/3 for 4 predictions
  - Avg NLL (base e) = 1.242 Perplexity = 3.464 ← geometric mean of denominators

LM Evaluation - Perplexity



LM Evaluation - Perplexity

- The perplexity of modern language models have consistently been going down.



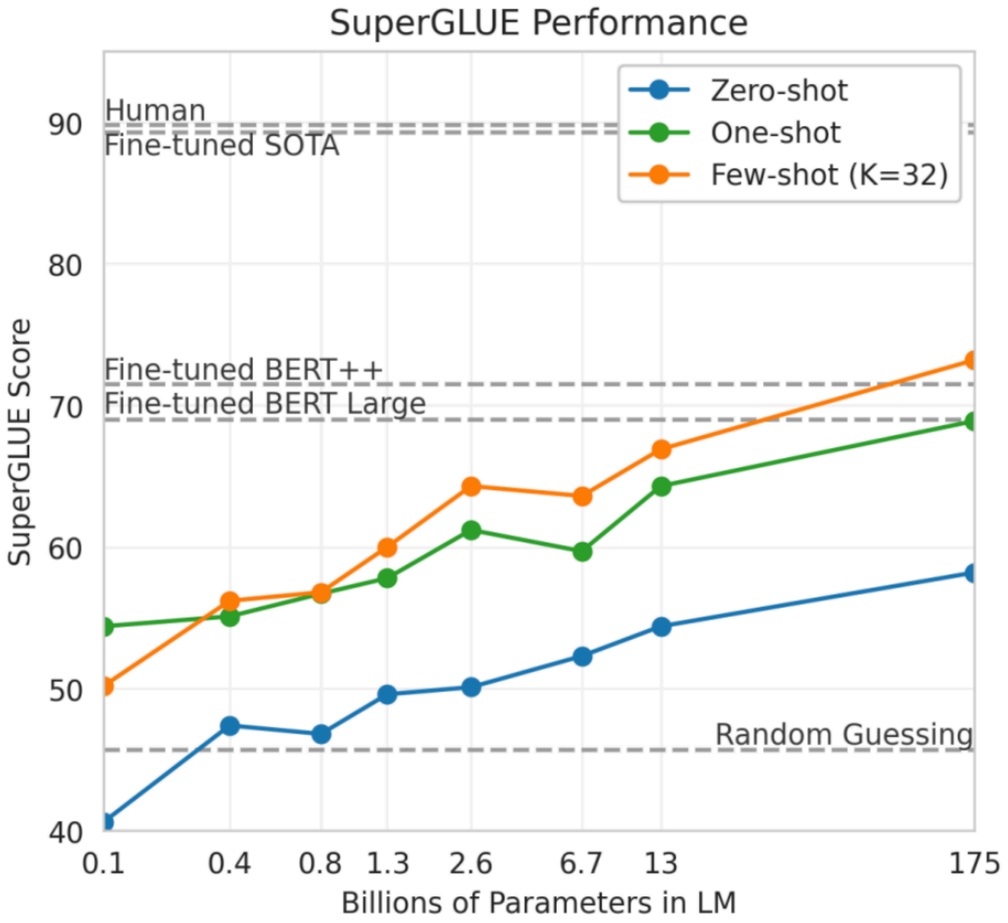
GPT-3

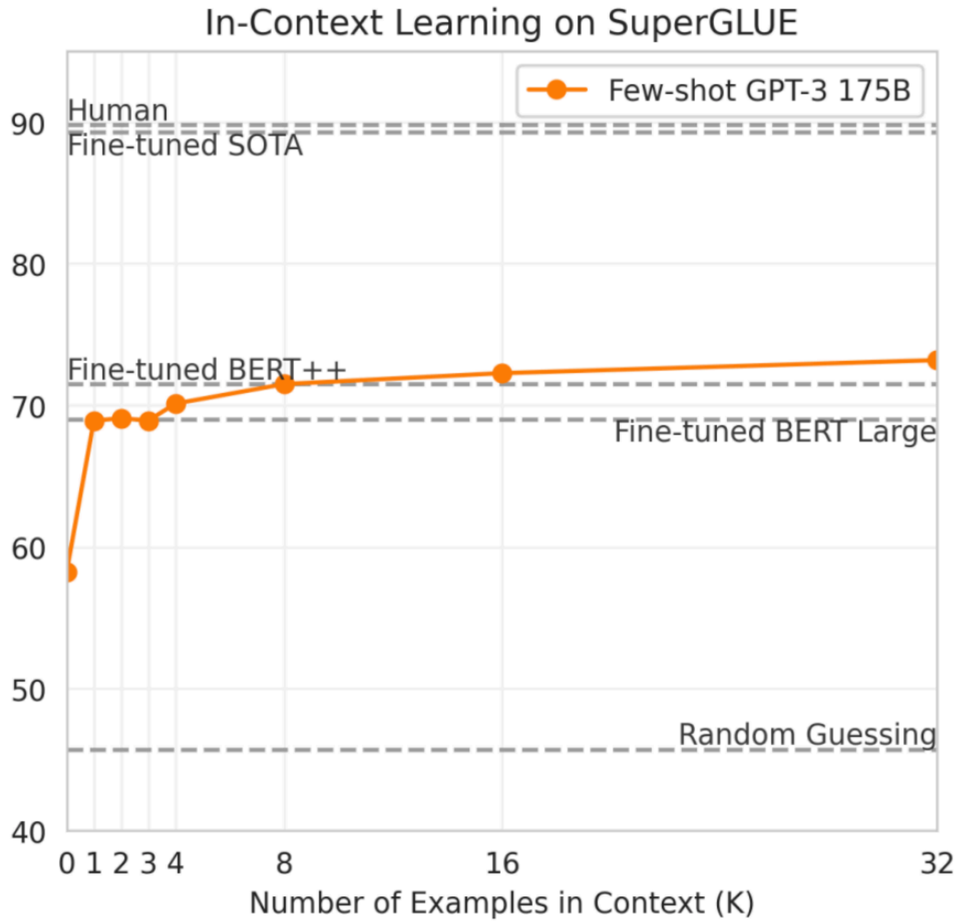
- GPT-2 but even larger: 1.3B (GPT2) —> 175B (GPT3) parameter models
- Sparse factorizations of the attention matrix to reduce computing time and memory use. context window is set to 2048 tokens.
- Data: filtered Common Crawl (410B tokens downsampled x0.44) + WebText dataset (19B x2.9) + two Internet-based book corpora (12Bx1.9, 55Bx0.43) + English Wiki (3B upsampled x3.4)

https://twitter.com/cocoweixu/status/1285727605568811011

Brown et al. (2020)

GPT-3 Performance





SuperGLUE benchmark - Wang et al. (2019)

GPT-3

- GPT-2 but even larger: 1.3B —> 175B parameter models

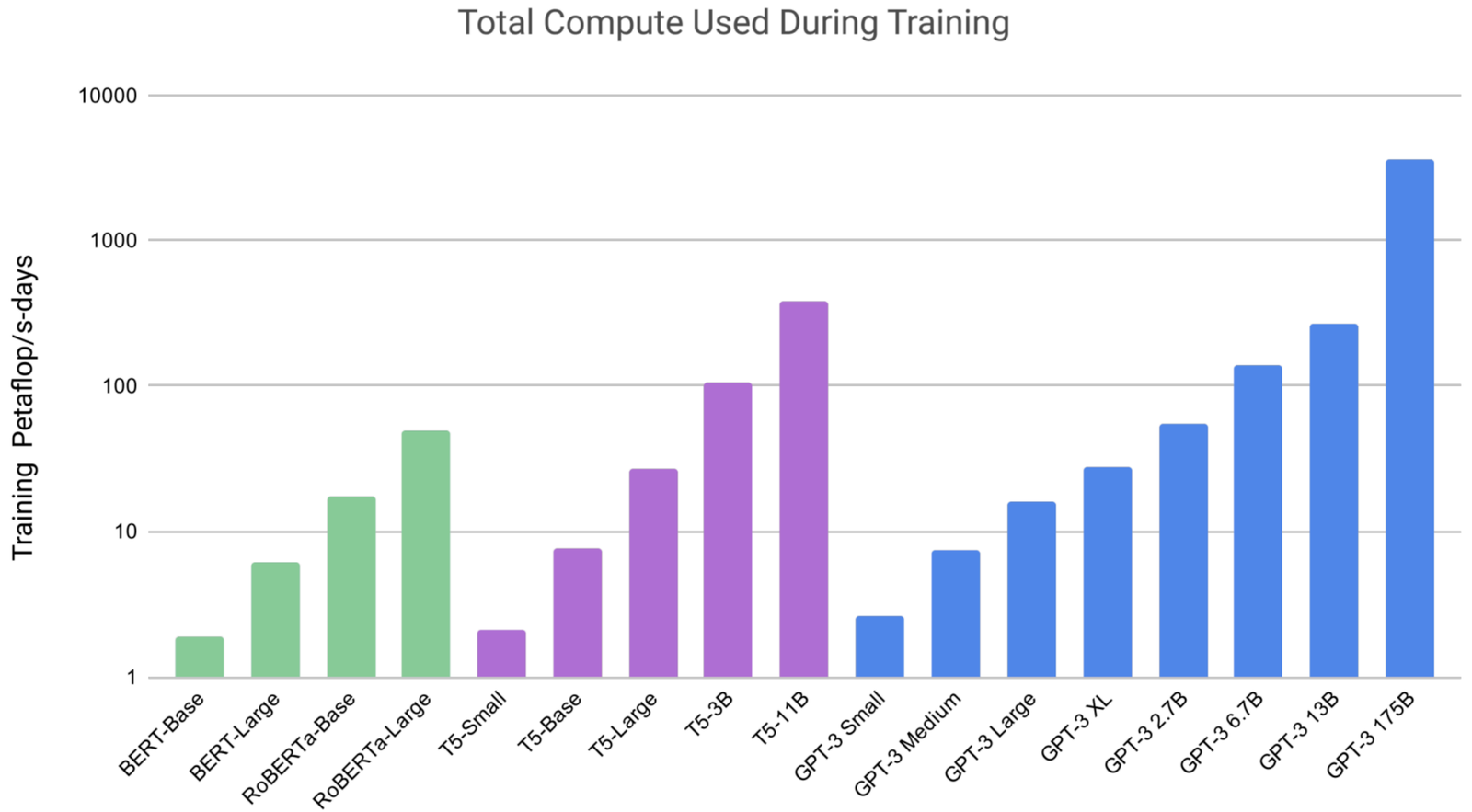
Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

- Trained on 570GB of Common Crawl
- 175B parameter model’s parameters alone take >400GB to store (4 bytes per param). Trained in parallel on a “high bandwidth cluster provided by Microsoft”

Brown et al. (2020)

GPT-3 Pre-training Cost

- Trained on Microsoft Azure, estimated to cost $4~10M (1000x BERT-large)

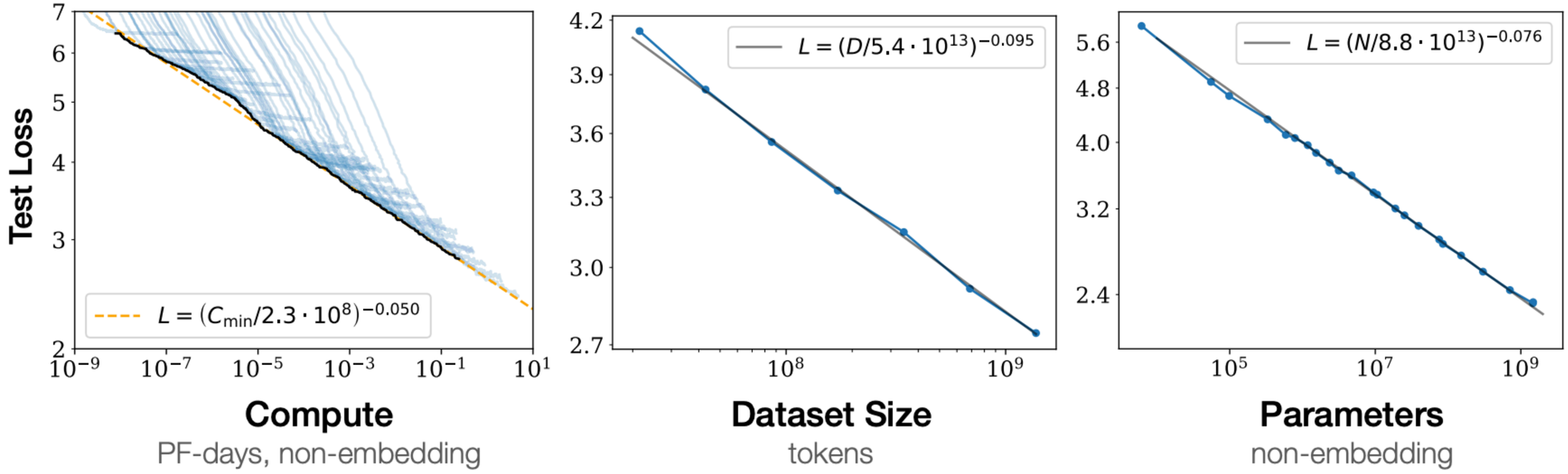


1 petaflop/s-day is equivalent to 8 V100 GPUs at full efficiency of a day

Brown et al. (2020)

GPT Scaling Laws

- With more compute, larger models get further down the loss “frontier”
- Building a bigger model (increasing compute) will decrease test loss!



petaflop (10^{20})/s-days

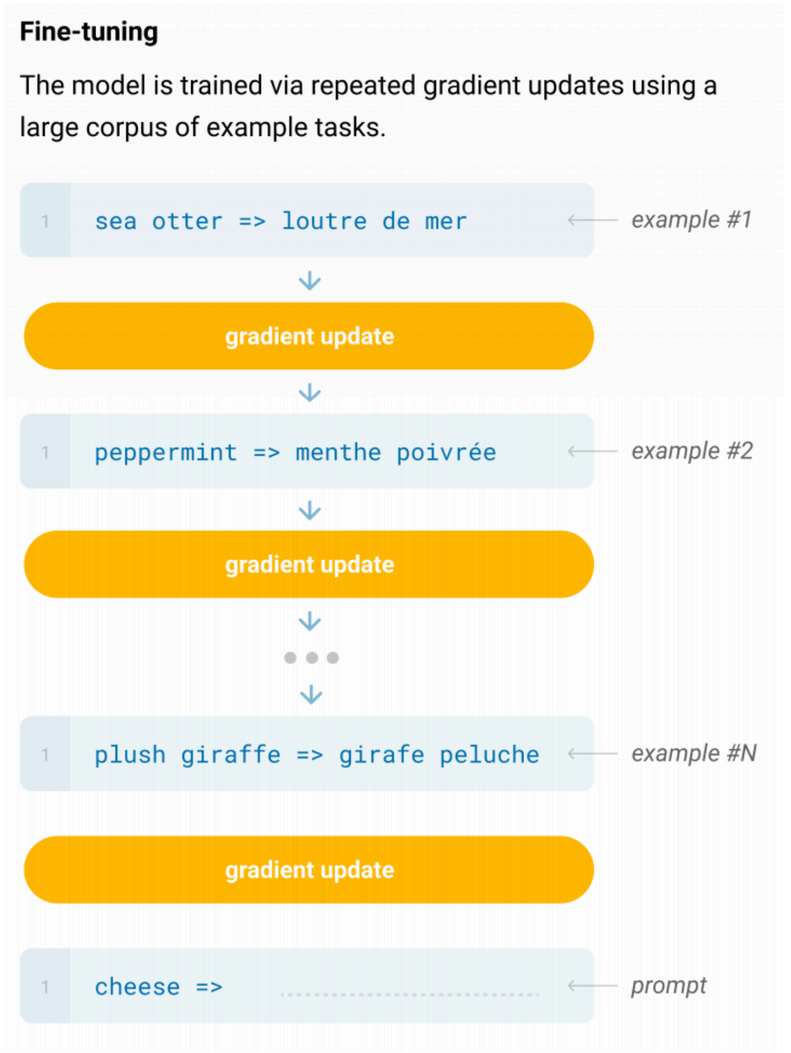
Kaplan et al. (2020)

Prompting

GPT-3: A Paradigm Shift!

- Until GPT-3, fine-tuning on task-specific labeled datasets was the standard way to use pre-trained models (e.g., BERT, T5, GPT-2) for downstream tasks (e.g., text classification, NER, etc.)

- Fine-tuning requires computing gradients and updating model parameters — computationally expensive for a 175B GPT-3 model



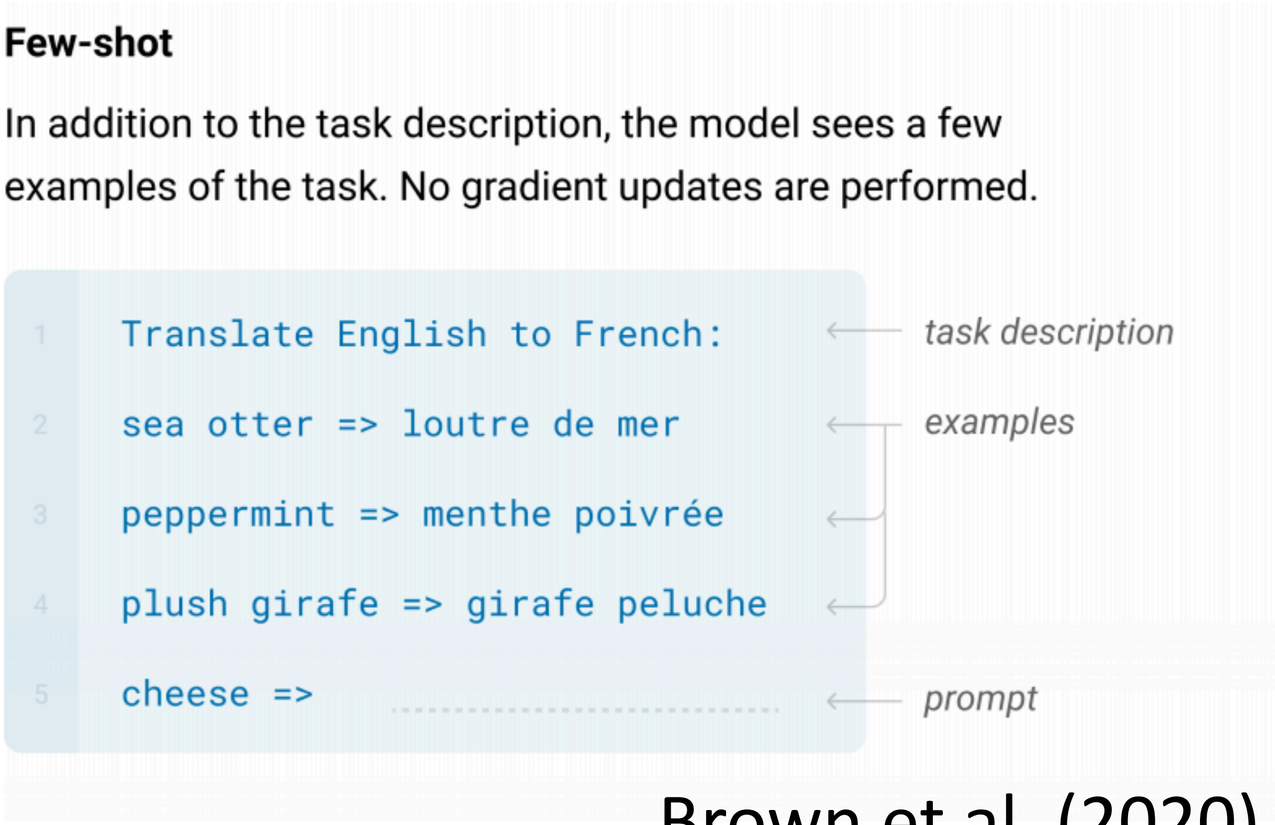
Brown et al. (2020)

GPT-3: Few-shot Prompting

- Model is frozen. No parameter updates.
  (i.e., only a forward pass, no backward pass)

- Prompt: task instruction plus a few demonstrations (training examples)

- This is also called:
  in-context learning



Brown et al. (2020)

GPT-3: Few-shot Prompting

- Model is frozen and is given a few demonstrations.

Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // _____

Circulation revenue has increased by 5% in Finland. // Finance

They defeated … in the NFC Championship Game. // Sports

Apple … development of in-house chips. // Tech

The company anticipated its operating profit to improve. // _____

Brown et al. (2020), Schick et al. (2020), Schick and Schütze (2021)

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- "in-context learning" - unlike conventional machine learning in that there's no optimization of any parameters.

Brown et al. (2020), Schick et al. (2020), Schick and Schütze (2021)

GPT-3: Few-shot Prompting

- Model is frozen and is given a few demonstrations.

Circulation revenue has increased by 5% in Finland. // Positive

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The company anticipated its operating profit to improve. // _____

- "in-context learning" - unlike conventional machine learning in that there's no optimization of any parameters.
- Model "learns" by conditioning on a few examples of the task.

Brown et al. (2020), Schick et al. (2020), Schick and Schütze (2021)

Prompting

- Zero-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}}(\mathbf{y} \mid \mathbf{i}, \mathbf{x})$$

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← i, task description
2  cheese =>                           ← x
```

Brown et al. (2020)

Prompting

- Few-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\text{LM}}} \left(\mathbf{y} \mid \mathbf{i}, \underbrace{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_K, \mathbf{y}_K)}_{\text{Few-shot examples}}, \mathbf{x} \right)$$

- Zero-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\text{LM}}} (\mathbf{y} \mid \mathbf{i}, \mathbf{x})$$

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1  Translate English to French:        ← i, task description
2  cheese =>                           ← x
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← i, task description
2  sea otter => loutre de mer          ← examples (x_1, y_1)
3  peppermint => menthe poivrée        ⋮
4  plush girafe => girafe peluche      ⋮
5  cheese =>                           ← x
```

Brown et al. (2020)

Prompting

- Few-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}} \left(\mathbf{y} \mid \mathbf{i}, \underbrace{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_K, \mathbf{y}_K)}_{\text{Few-shot examples}}, \mathbf{x} \right)$$

- Zero-shot Prompt:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p_{\theta_{\mathrm{LM}}} (\mathbf{y} \mid \mathbf{i}, \mathbf{x})$$

optional

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← x
```

Few-shot

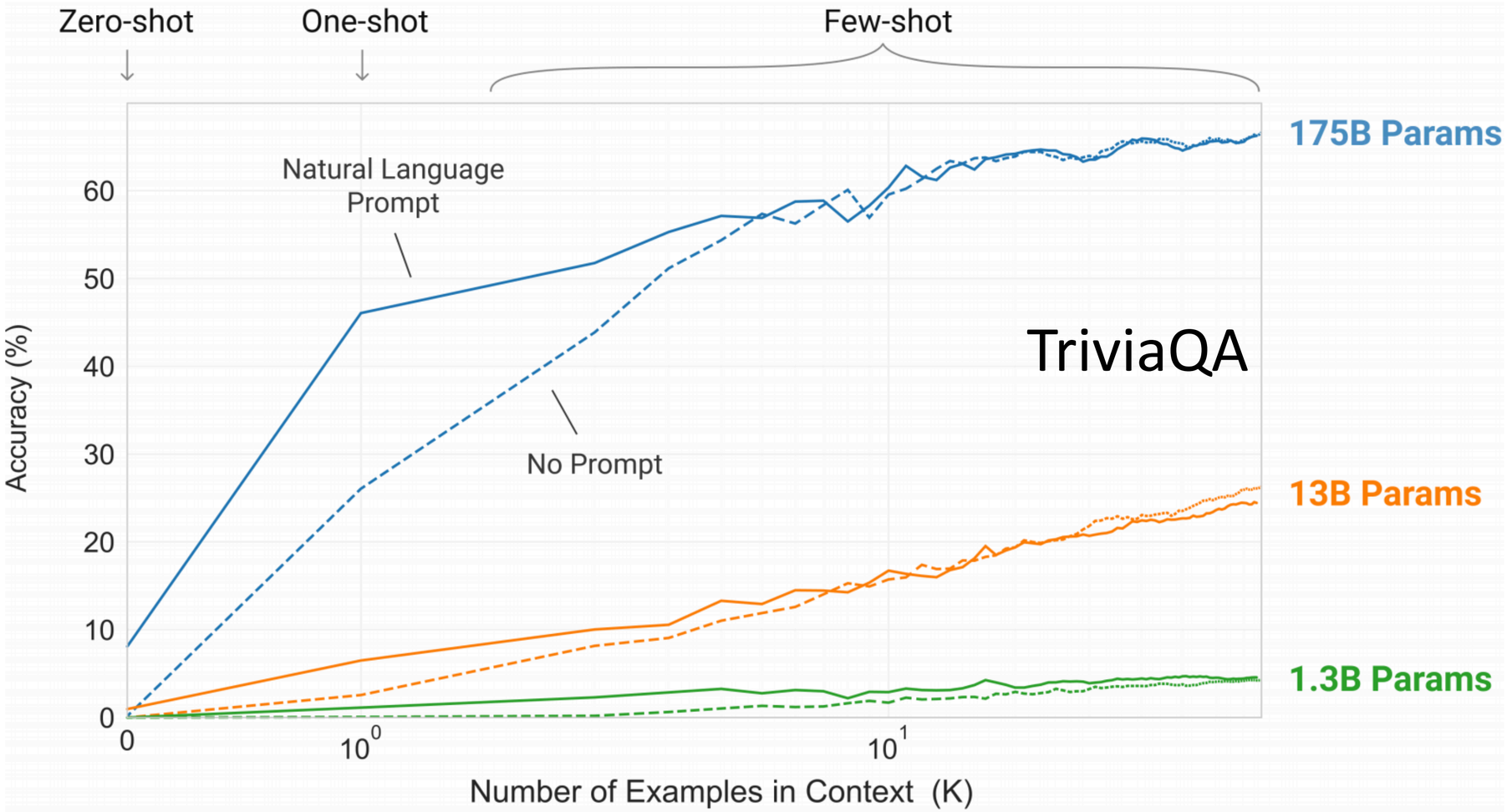
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4  plush girafe => girafe peluche      ⋮
5  cheese =>                           ← x
```

Brown et al. (2020)

GPT-3: Few-shot Learning

- **Key observation:** few-shot learning only works with the very largest models!



Brown et al. (2020)

TriviaQA

Context →	Q: ‘Nude Descending A Staircase’ is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

Figure G.34: Formatted dataset example for TriviaQA. TriviaQA allows for multiple valid completions.

MultiRC (multi-sentence)

Sent 1: The hijackers attacked at 9:28.
Sent 2: While traveling 35,000 feet above eastern Ohio, United 93 suddenly dropped 700 feet.
Sent 3: Eleven seconds into the descent, the FAA's air traffic control center in Cleveland received the first of two radio transmissions from the aircraft.
Sent 4: During the first broadcast, the captain or first officer could be heard declaring "Mayday" amid the sounds of a physical struggle in the cockpit.
Sent 5: The second radio transmission, 35 seconds later, indicated that the fight was continuing.
Sent 6: The captain or first officer could be heard shouting: "Hey get out of here-get out of here-get out of here."
Sent 7: On the morning of 9/11, there were only 37 passengers on United 93-33 in addition to the 4 hijackers.
Sent 8: This was below the norm for Tuesday mornings during the summer of 2001.
Sent 9: But there is no evidence that the hijackers manipulated passenger levels or purchased additional seats to facilitate their operation.
Sent 10: The terrorists who hijacked three other commercial flights on 9/11 operated in five-man teams.
Sent 11: They initiated their cockpit takeover within 30 minutes of takeoff.
Sent 12: On Flight 93, however, the takeover took place 46 minutes after takeoff and there were only four hijackers.

Question: Which two factors were different between the three other hijacked planes and United 93?
the day of the takeover
A) The amount of time that passed before the takeover started
B)* United 93 took longer and had less hijackers
C) The airline operating the planes
D) The weather and fuel used by the airplane
E) The navigation system used by the planes
Reasoning needed: Discourse relation (contrast)
One needs to identify that the discourse marker *however* in Sent 12 indicates a contrast relation between Flight 93 and the flights mentioned in Sent 10. Also, *only* in Sent 12 indicates that the number of hijackers were fewer than in the contrasted other flights.

Question: What was below average for this particular day?
A) the number of passengers in the first class.
B)* the number of passengers on board.
C) the number of hijackers
D) the amount of air traffic in the skies
E) the temperature
Reasoning needed: Event coreference
One needs to identify that *This* in Sent 8 co-refers to (event of) number of passengers in Sent 7. Note that Sent 12 contains *only four hijackers* and understanding that *only* indicates a smaller number of entities than expected (as in previous question), might mislead a system into believing that (C) is the correct answer.

GPT-3

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

- Sometimes very impressive, (MultiRC, ReCoRD), sometimes very bad
- Results on other datasets are equally mixed — but still strong for a few-shot model!

Brown et al. (2020)

Scaling Laws

Wei Xu

Compute Cost

- For a fixed compute budget C, there is a tradeoff between N and D

- For example, training compute for a Transformer LM is approximately:

$$C \approx 6ND$$

$$(\text{FLOPs} \approx 6 \cdot \text{params} \cdot \text{tokens})$$

  - Model size **N**: number of parameters
  - Training dataset size **D**: number of tokens/examples
  - Training compute **C**: floating point operations (FLOPs)

Training LLMs under Resource Limits

- Typically, there is:
  - a limit on compute budget (number of GPUs × training time)
  - a limit on the amount of training data available
- To build a very large language model, we can only afford training once
- Q: What is the optimal model size (number of parameters) to train?

Scaling Laws

- How do we allocate the resources more optimally?
- → How model performance would change as resources increase?

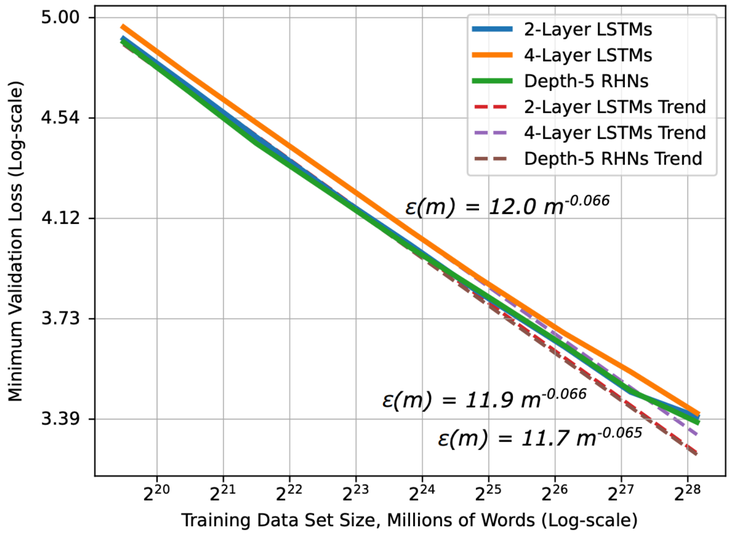
Performance = function (N, D, C)

  - Model size **N**: number of parameters
  - Training dataset size **D**: number of tokens/examples
  - Training compute **C**: floating point operations (FLOPs)

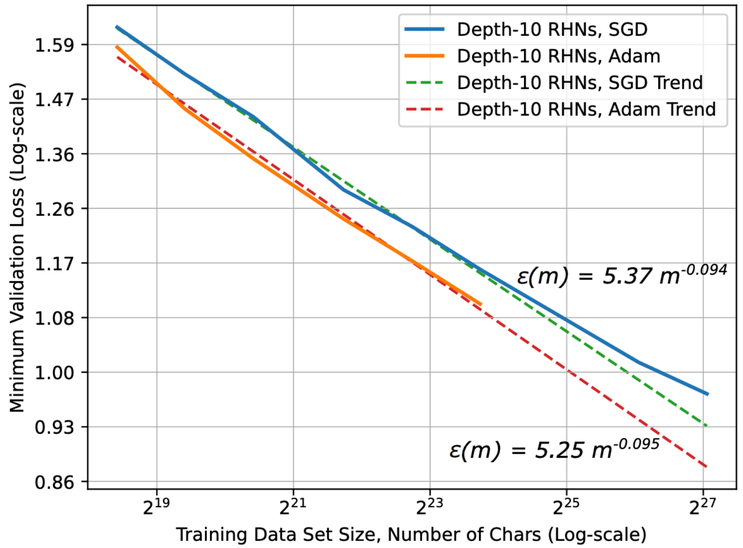
- Other factors (e.g., model architecture, hyperparameters) fixed/tuned

Baidu Scaling Laws

- Predates modern LLM scaling laws
- **Power-law** relationships with dataset and model size
- Observed across diverse tasks and neural network architectures



Machine Translation
(seq2seq - Luong et a. 2015)



Character Language Models
(RHN - Zilly et al. 2017)

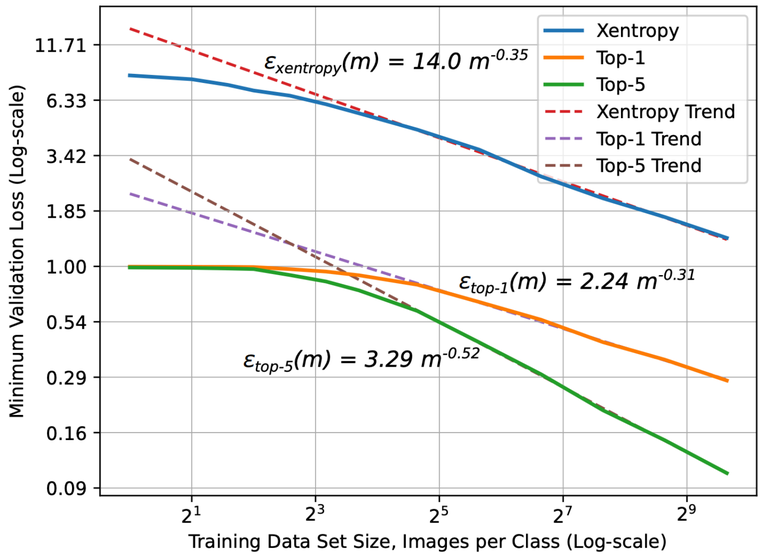
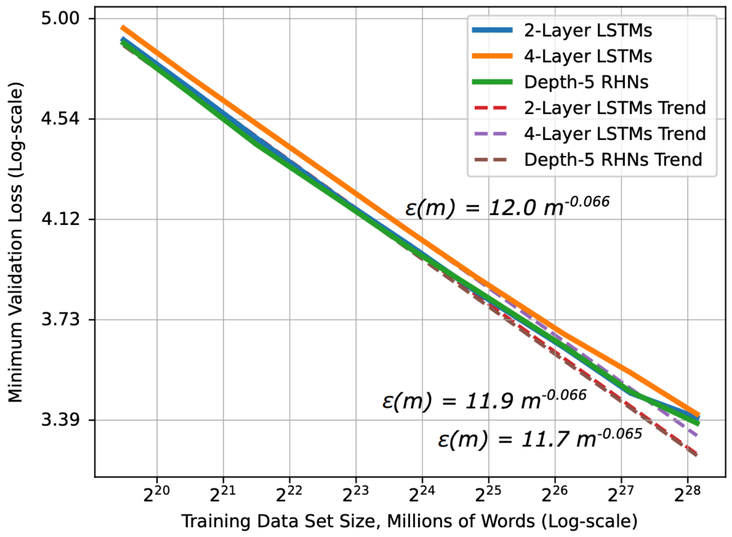


Image Classification
(ResNets - He et al. 2016)

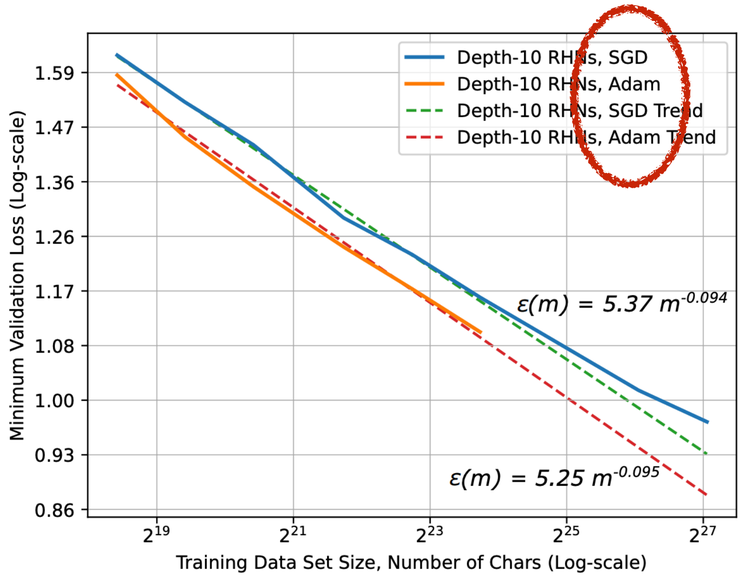
Hestness et al. (2017)

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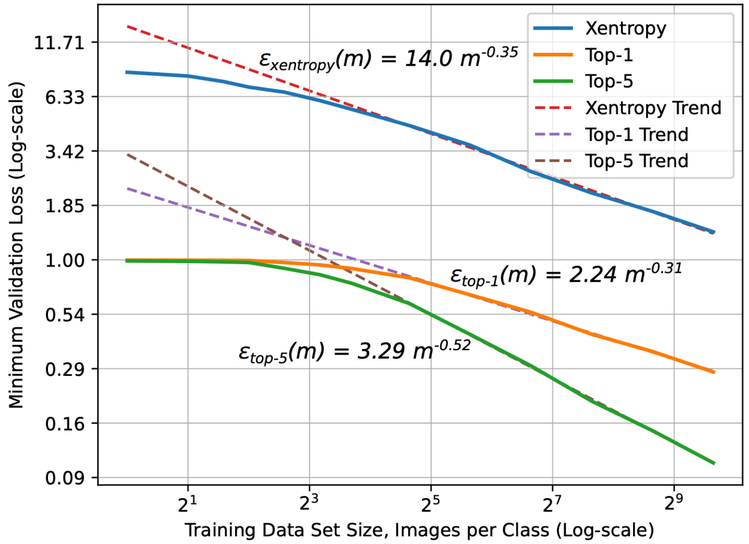
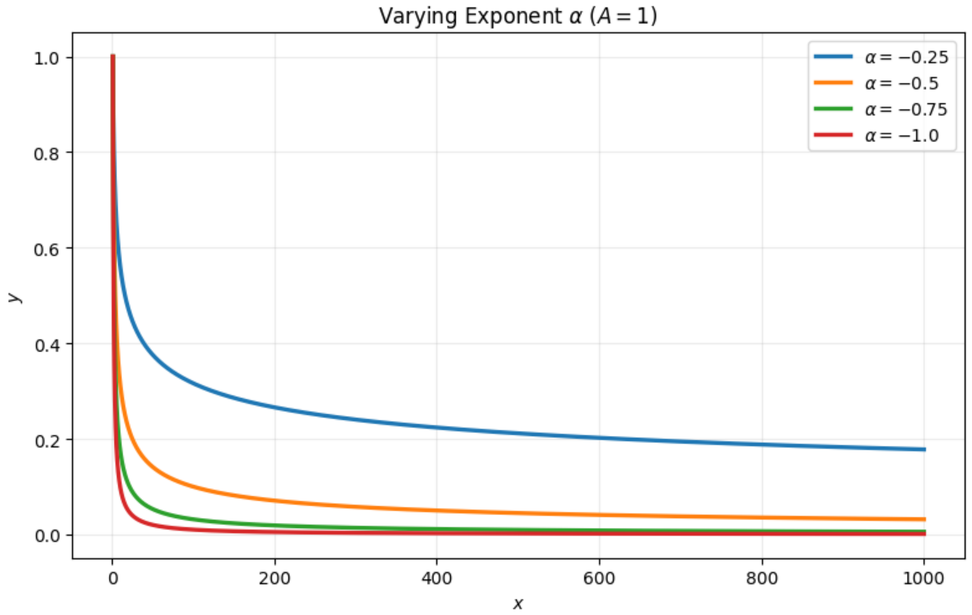


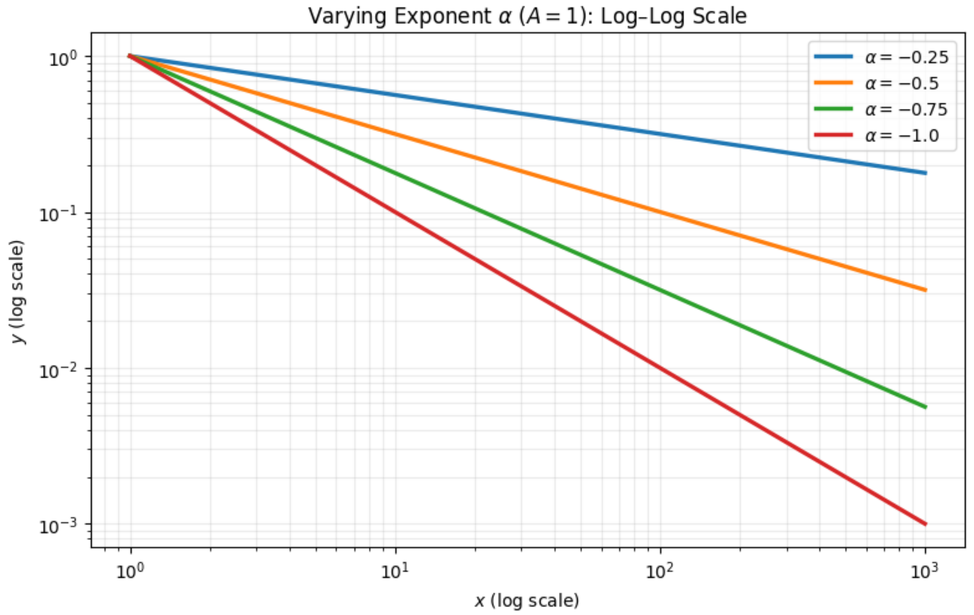
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(ResNets - He et al. 2016)

Hestness et al. (2017)

Power Law

$$y = Ax^{\alpha} \quad \Longrightarrow \quad \log y = \underbrace{\log A}_{\text{intercept}} + \underbrace{\alpha}_{\text{slope}} \log x$$

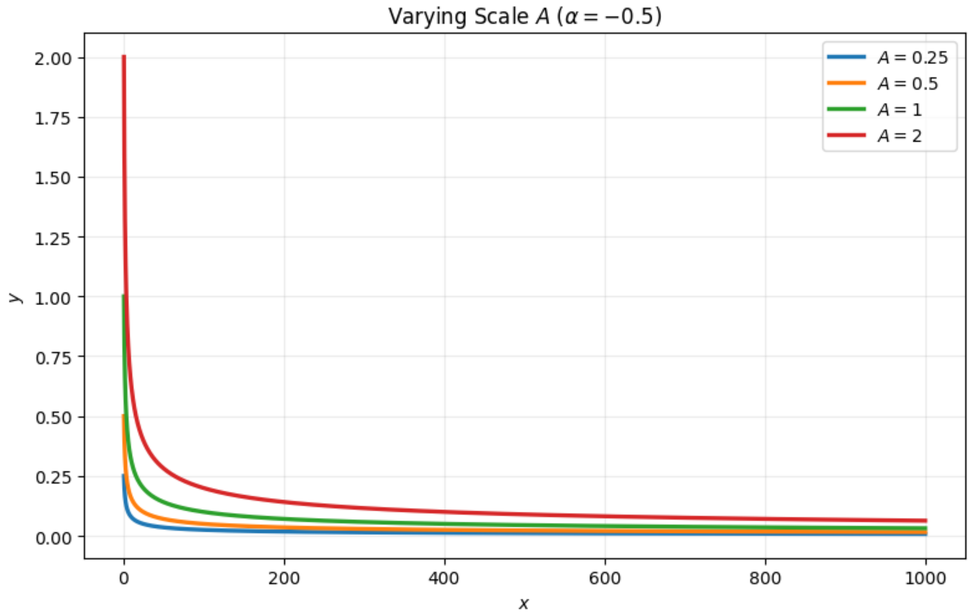


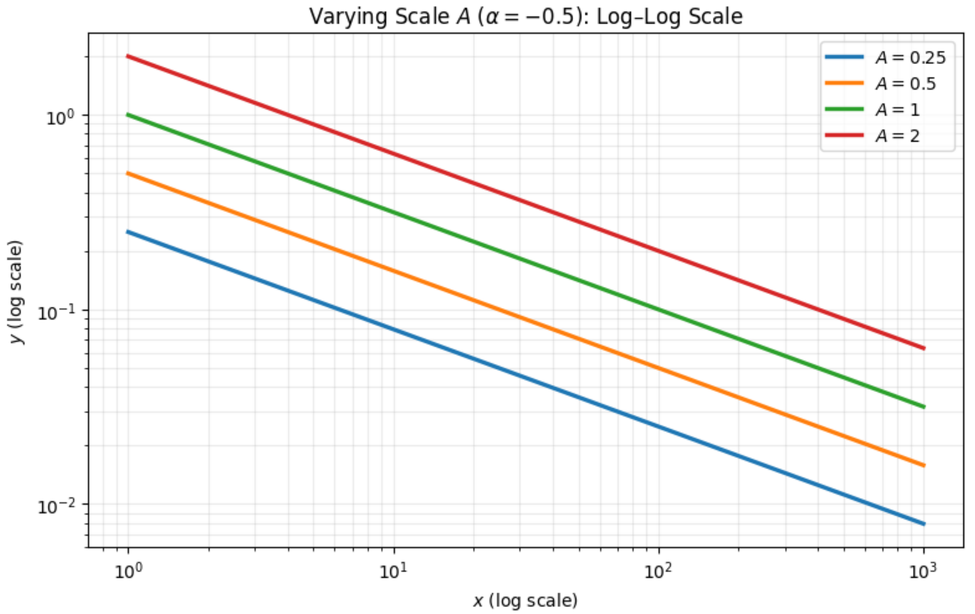


- More negative α means faster improvement (lower loss/error) with scale

Power Law

$$y = Ax^{\alpha} \quad \Longrightarrow \quad \log y = \underbrace{\log A}_{\text{intercept}} + \underbrace{\alpha}_{\text{slope}} \log x$$

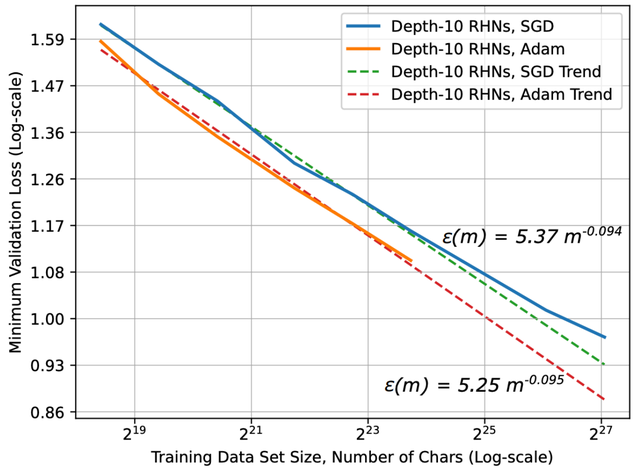




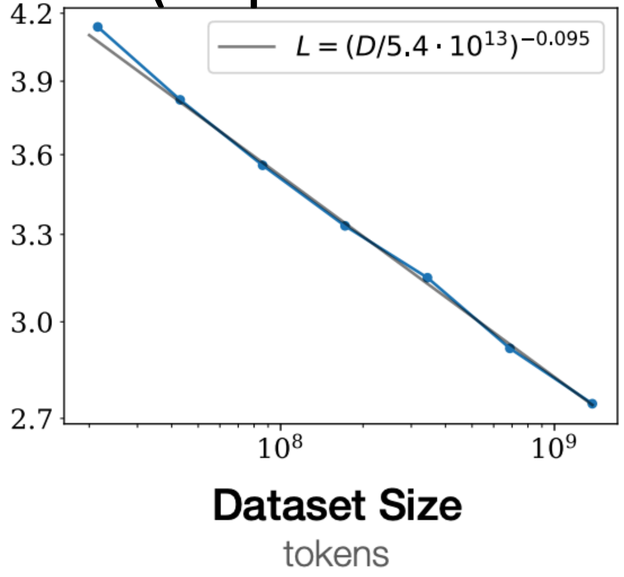
- From Baidu's scaling law paper: improved model architectures and optimizers can improve the power-law intercept, but not the exponent
- i.e., models for a single domain show the same learning curve steepness.

Dataset Scaling Laws

Baidu (Hestness et al. 2017)

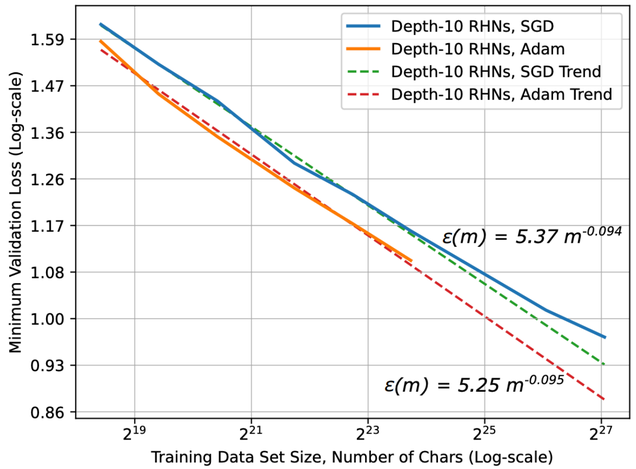


OpenAI (Kaplan et al. 2020)



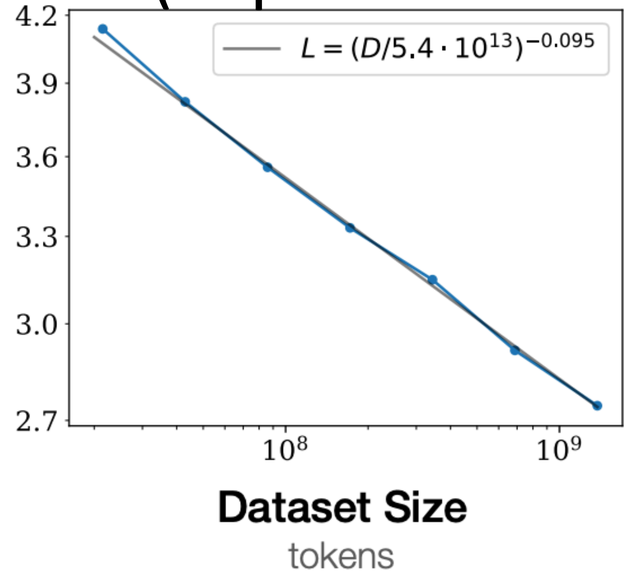
Dataset Scaling Laws

Baidu (Hestness et al. 2017)



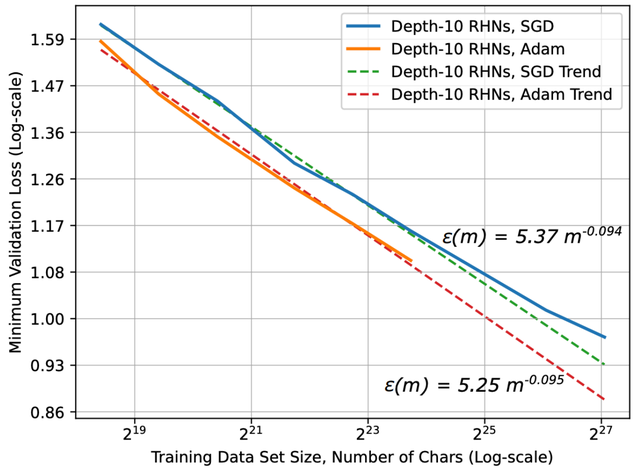
- General power law: $y = Ax^{\alpha}$

OpenAI (Kaplan et al. 2020)

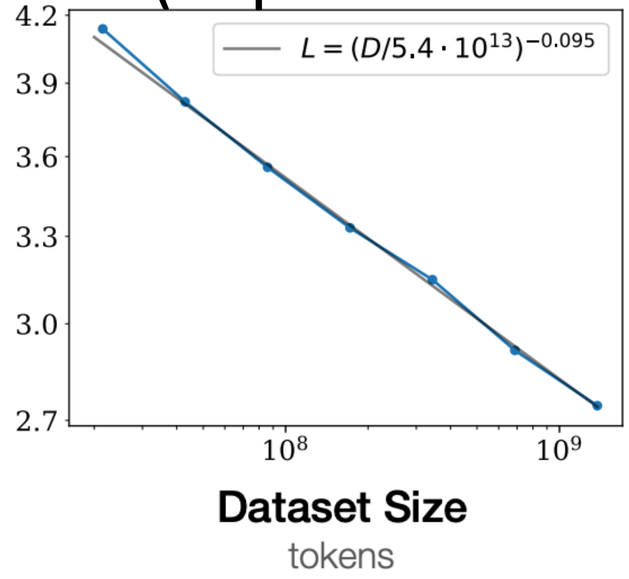


Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)

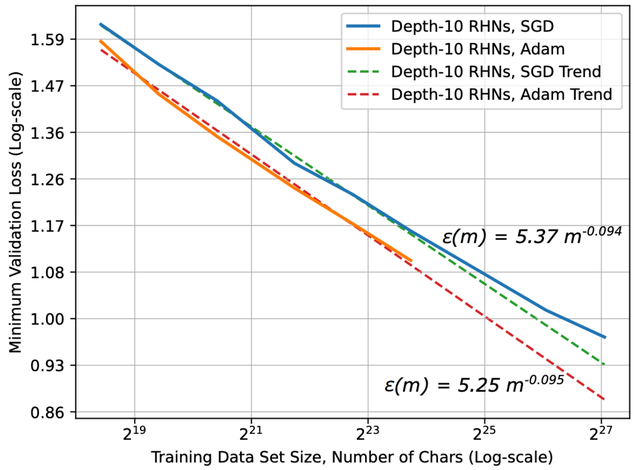


- General power law: $y = Ax^{\alpha}$
- Dataset scaling law:

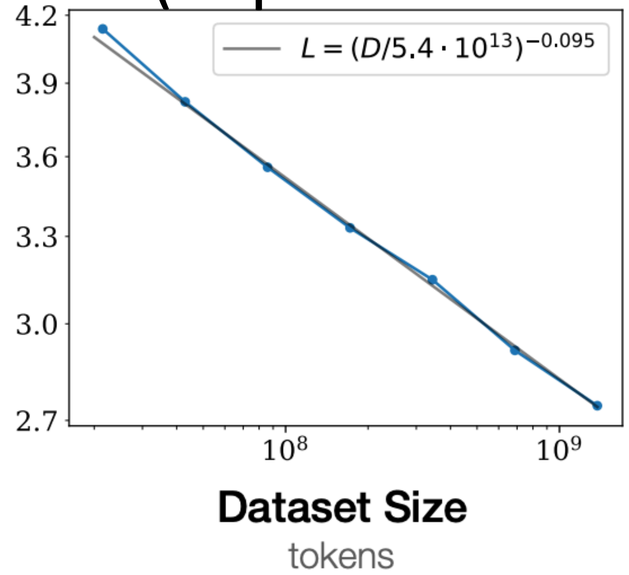
$$L(D) \approx \left(\frac{D}{D_c}\right)^{-\alpha_D}$$

Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)



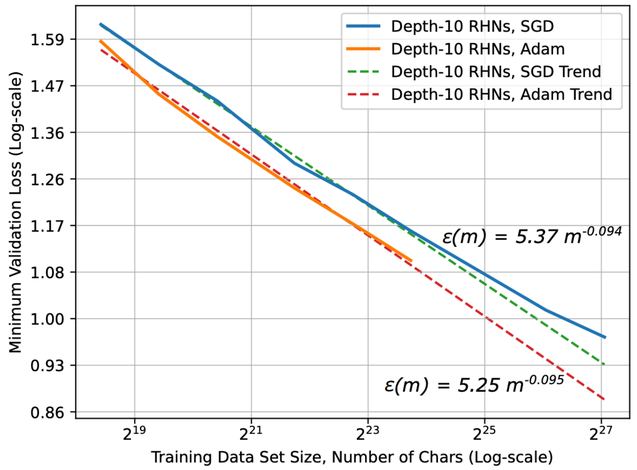
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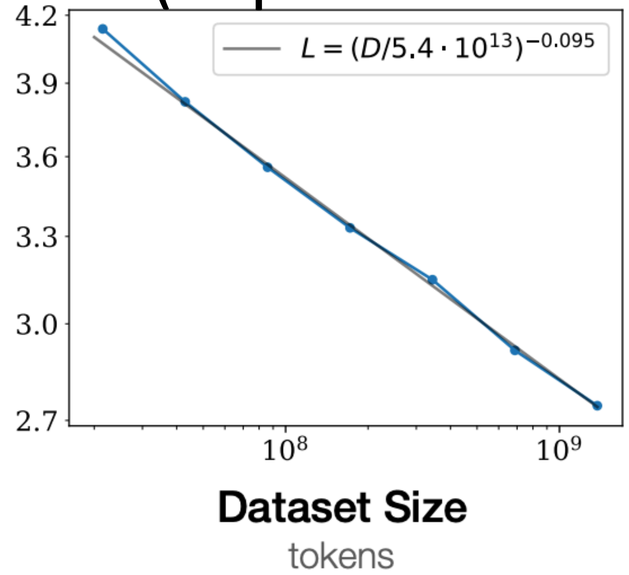
$$= \underbrace{D_c^{\alpha_D}}_{A} \underbrace{D^{-\alpha_D}}_{x^{\alpha}}$$

Dataset Scaling Laws

Baidu (Hestness et al. 2017)



OpenAI (Kaplan et al. 2020)



- General power law: $y = Ax^{\alpha}$
- Dataset scaling law:

$$L(D) \approx \left(\frac{D}{D_c}\right)^{-\alpha_D}$$

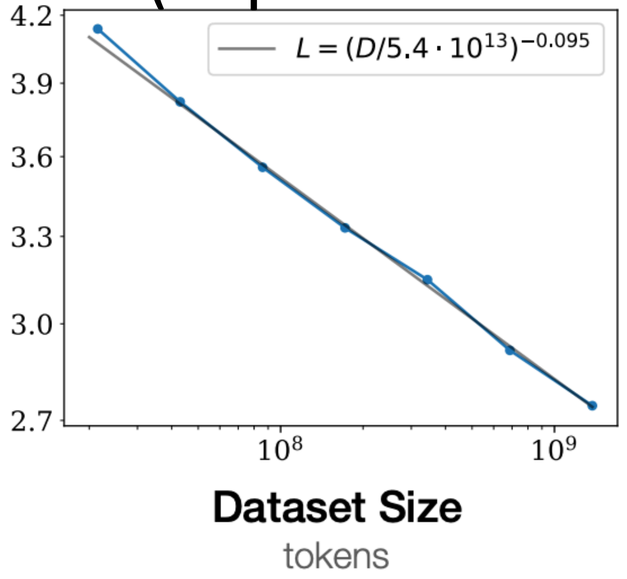
$$= \underbrace{D_c^{\alpha_D}}_{A} \underbrace{D^{-\alpha_D}}_{x^{\alpha}}$$

$$\propto D^{-\alpha_D}$$

Dataset Scaling Laws

- The exponent captures the fundamental scaling behavior:
  i.e. how quickly loss decreases as D increases?

OpenAI (Kaplan et al. 2020)

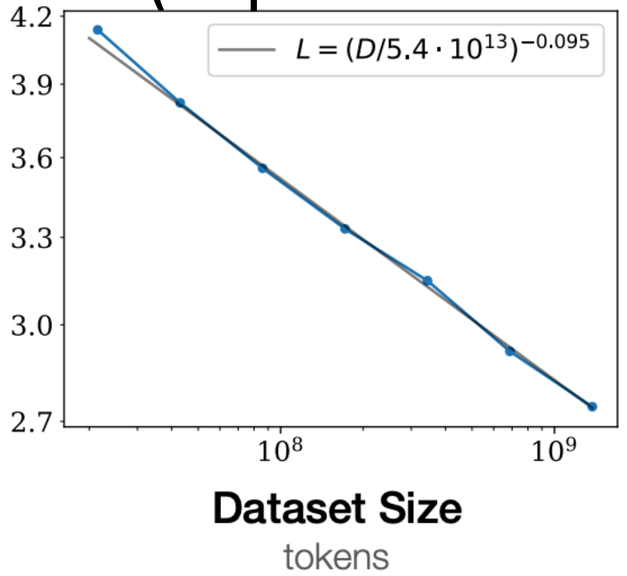


Dataset Scaling Laws

- The exponent captures the fundamental scaling behavior:
  i.e. how quickly loss decreases as D increases?

$$L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D} = \left(\frac{D}{D_c}\right)^{-\alpha_D} = D_c^{\alpha_D} D^{-\alpha_D} \propto D^{-\alpha_D}$$

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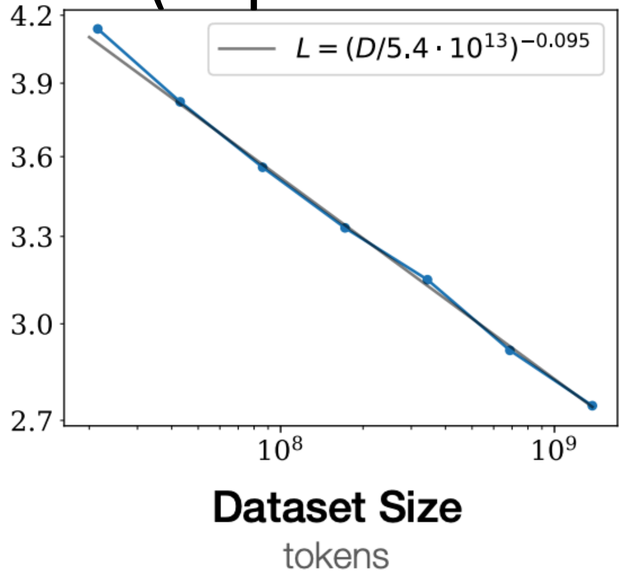


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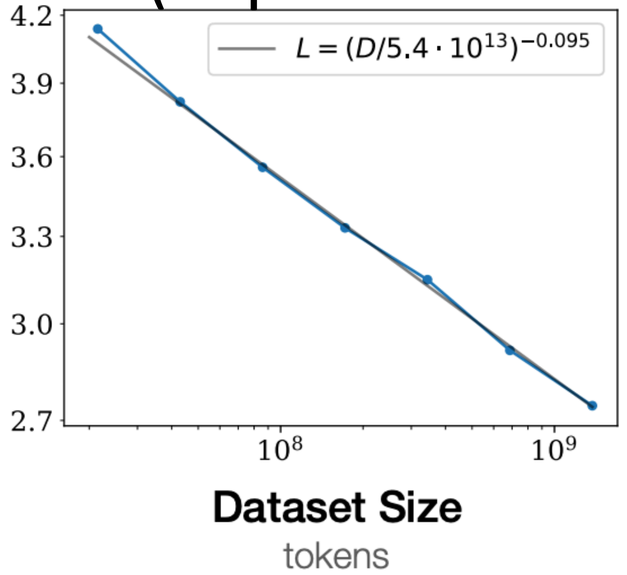
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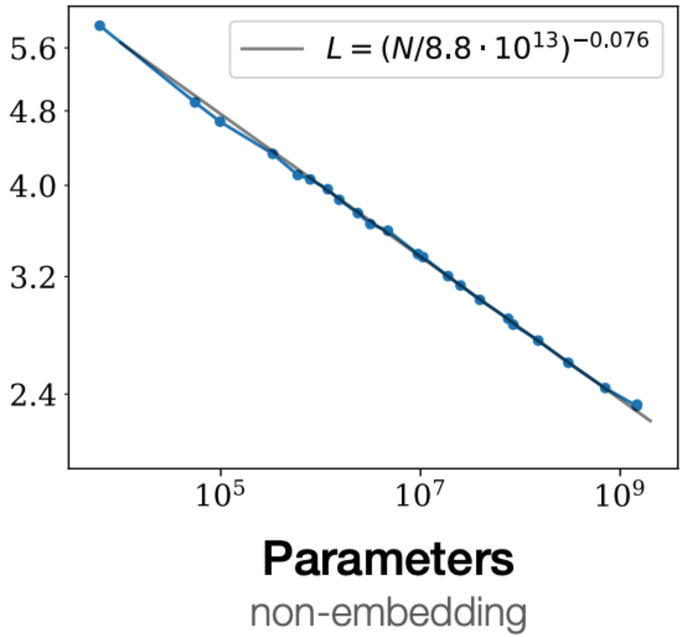


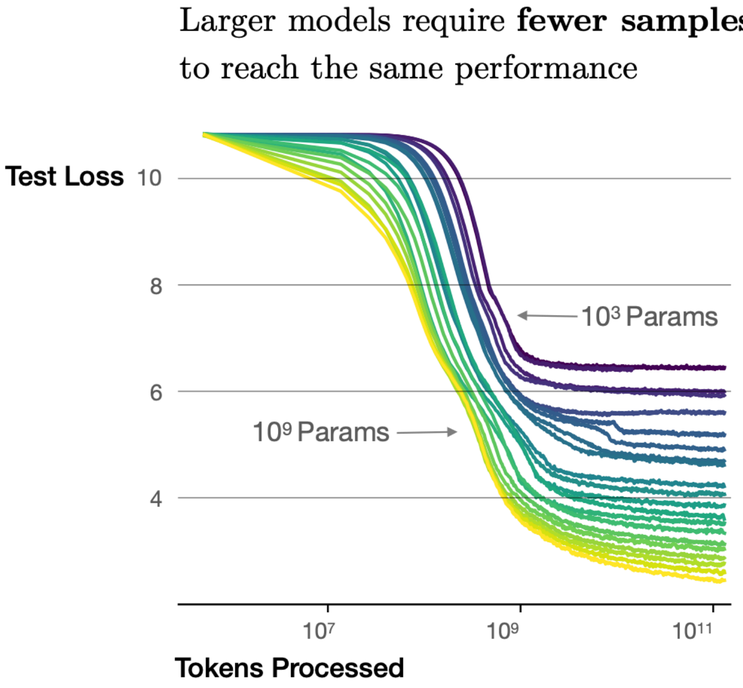
- For example, $L(D) \propto D^{-0.095}$
- Increasing D by 10x, will reduce loss by ~20%

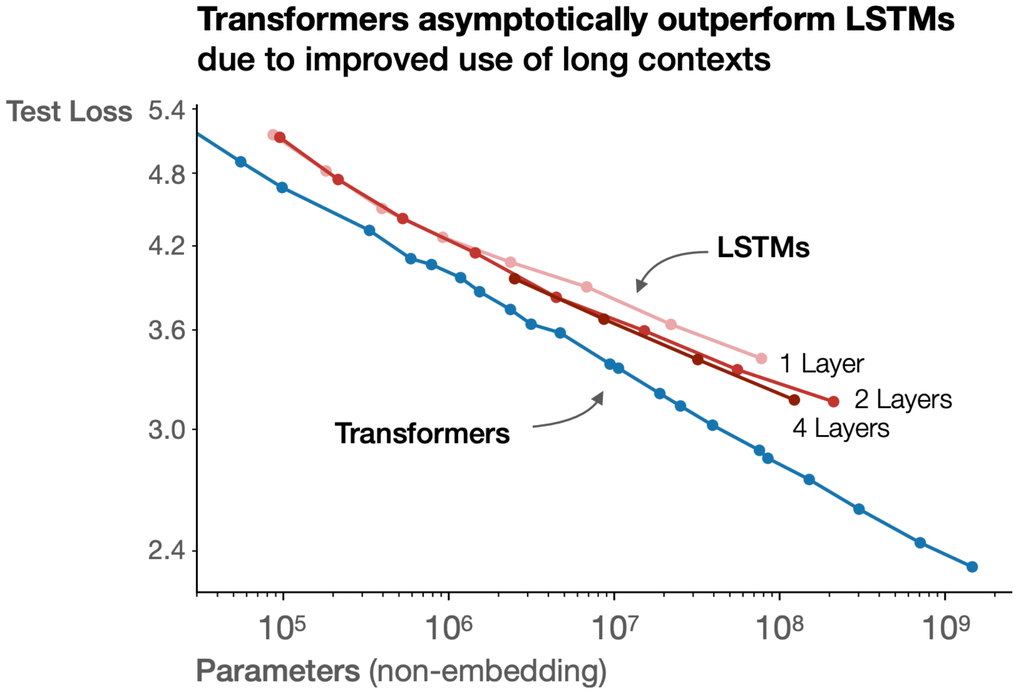
$$\frac{L(10D)}{L(D)} = \frac{(10D)^{-0.095}}{D^{-0.095}} = 10^{-0.095} \approx 0.803$$

Model Size Scaling Laws

- Similar to the dataset scaling laws

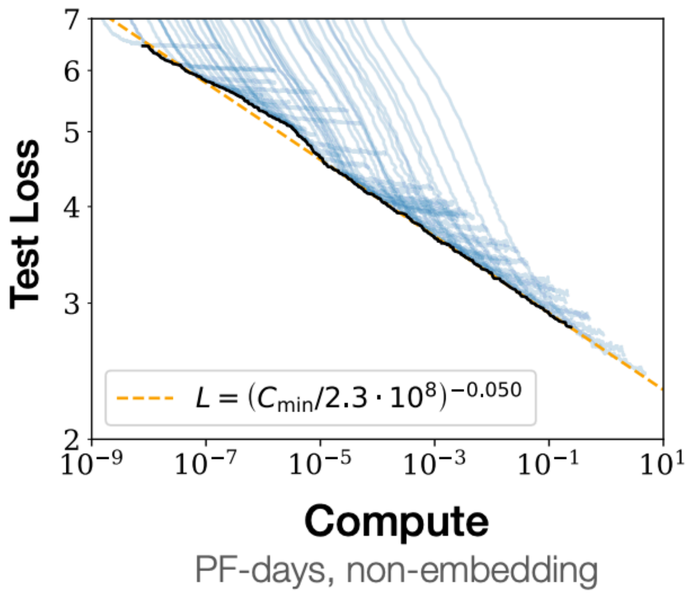






OpenAI (Kaplan et al. 2020)

Compute Scaling Laws

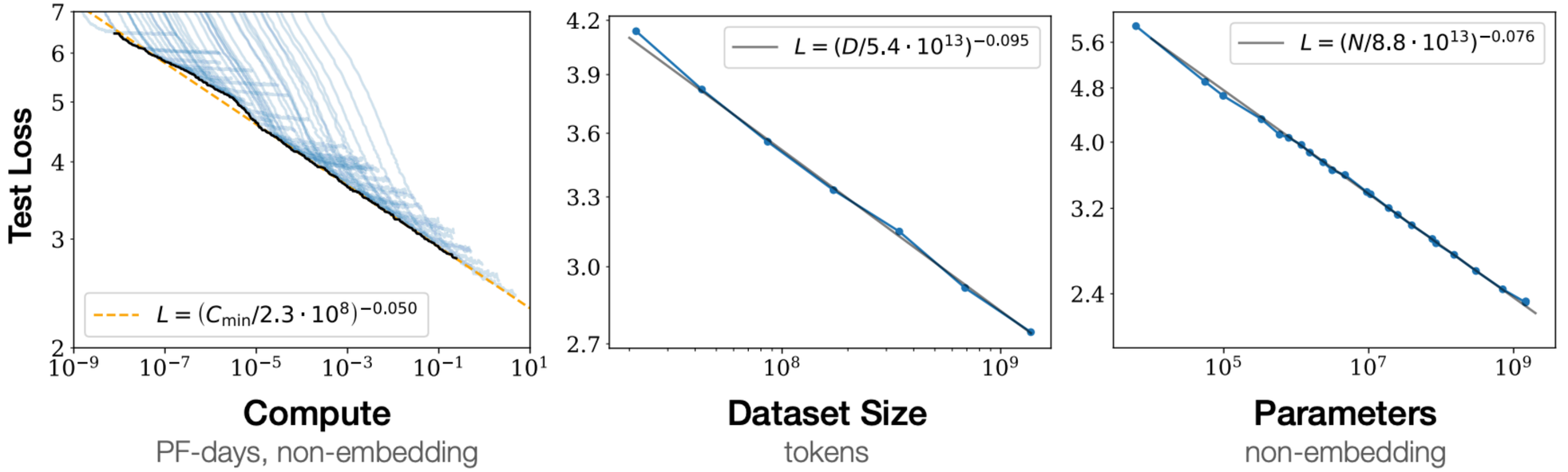


- Blue lines: model of different sizes N, trained on increasing amount of data D
- **Black** line - “compute-efficient frontier”: lowest test loss achieved by any model at a given compute budget
- Orange dashed line: use log-space regression to fit a power law

OpenAI (Kaplan et al. 2020)

GPT Scaling Laws

- With more compute, larger models get further down the loss “frontier”
- Building a bigger model (increasing compute) will decrease test loss!



$$L(C) \propto C^{-0.050}$$

$$L(D) \propto D^{-0.095}$$

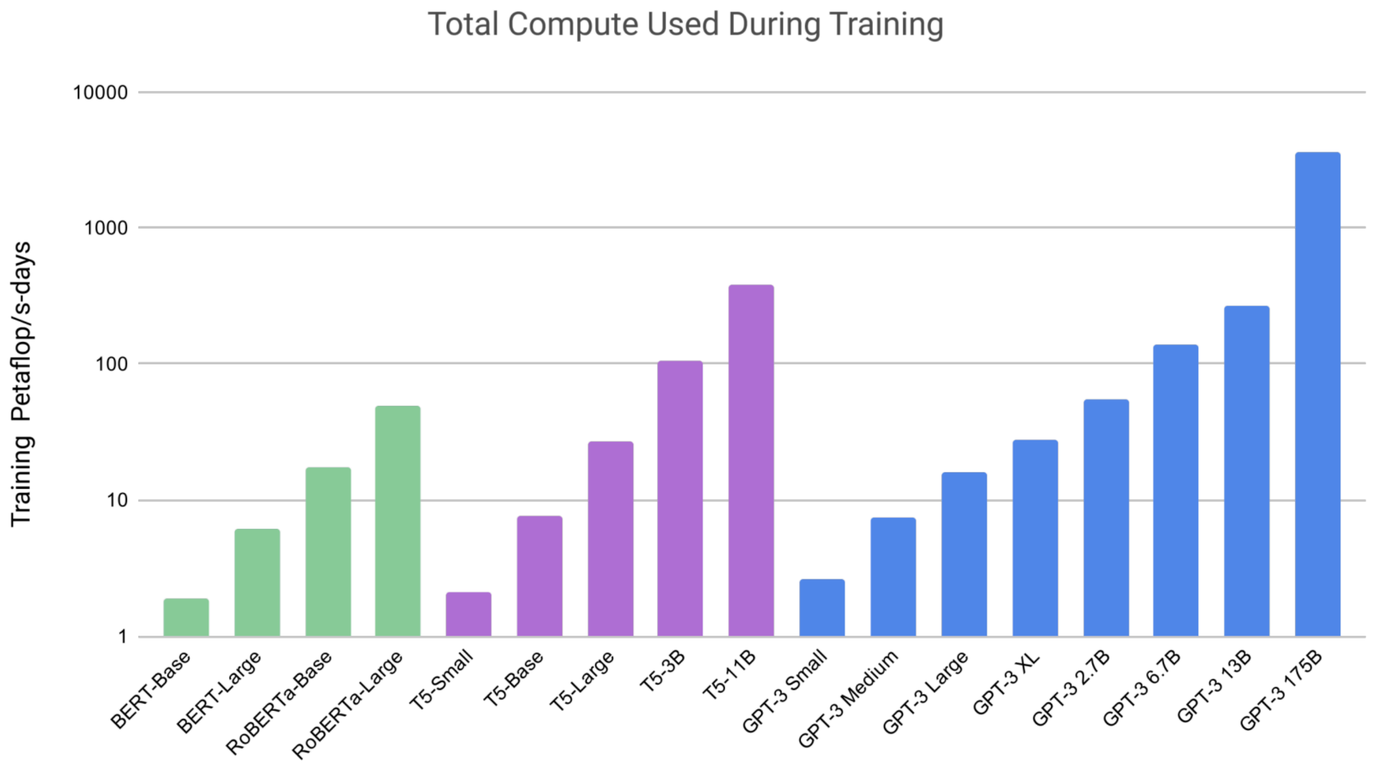
$$L(N) \propto N^{-0.076}$$

Kaplan et al. (2020)

GPT Scaling Laws

- For a fixed compute budget, Kaplan et al. (2020) argued for scaling model size much faster than training data (train bigger models!):

$$N_{\text{opt}} \propto C^{0.73}, \qquad D_{\text{opt}} \propto C^{0.27}$$

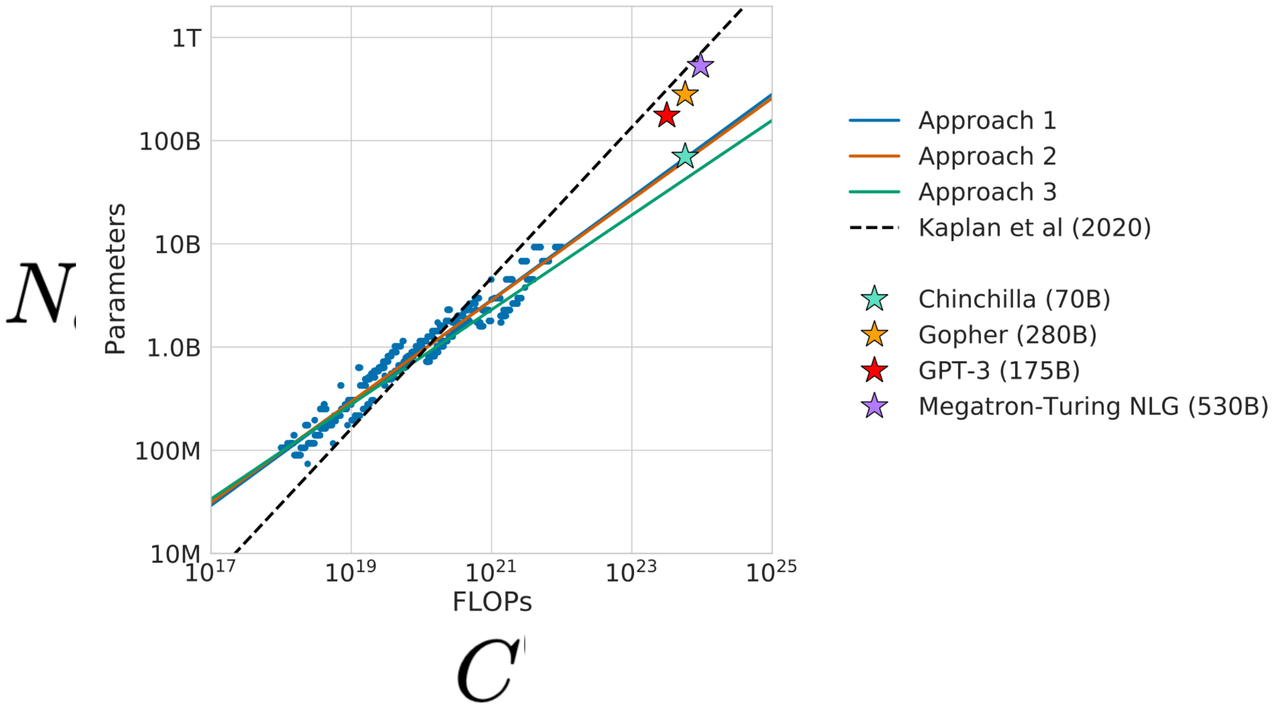


Kaplan et al. (2020)

Chinchilla Scaling Laws

- In contrast, Hoffmann (2022) argued for scaling model size and training data equally (training models on more data, avoid under-training):

$$N_{\text{opt}} \propto C^{0.50}, \qquad D_{\text{opt}} \propto C^{0.50}$$



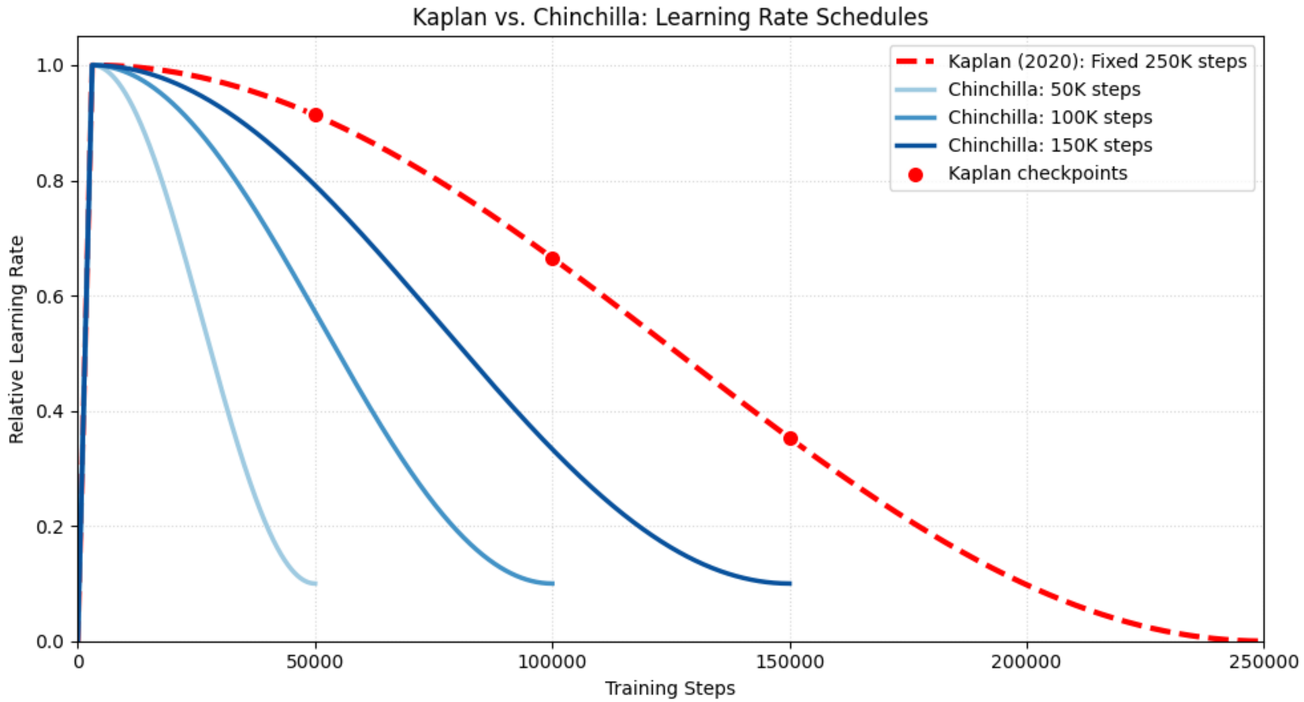
MMLU results:

Random	25.0%
Average human rater	34.5%
GPT-3 5-shot	43.9%
Gopher 5-shot	60.0%
***Chinchilla* 5-shot**	**67.6%**
Average human expert performance	*89.8%*

Hoffmann et al. (2022)

Chinchilla Scaling Laws

- Learning rate: adaptive schedule vs. Kaplan+ (2020)'s fixed schedule
- Model size: up to 70B vs. Kaplan+ (2020)'s 1.5B non-embedding parameters



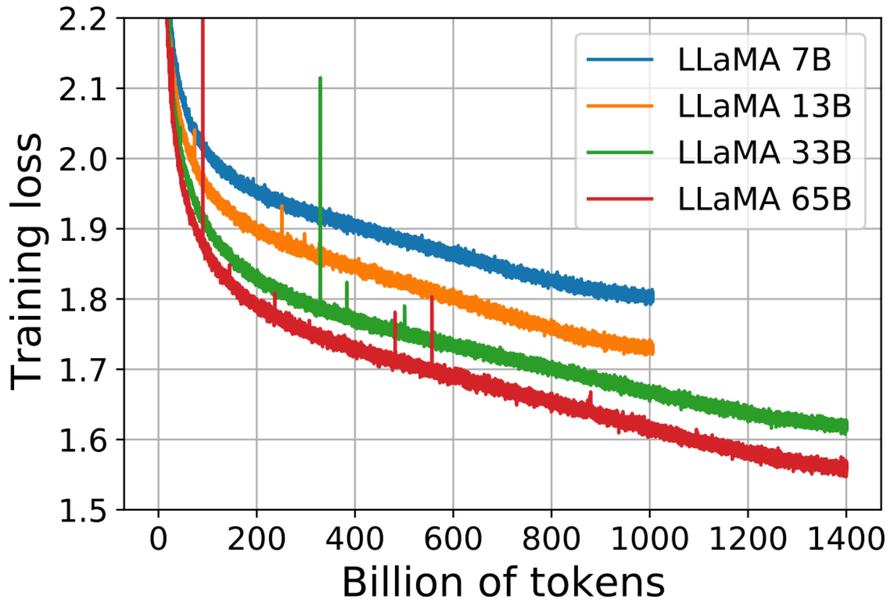
Hoffmann et al. (2022)

Llama (Preview)

- Besides scale dataset and model sizes for **training** compute budget, it is also important to consider the **inference (test-time)** compute budget

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

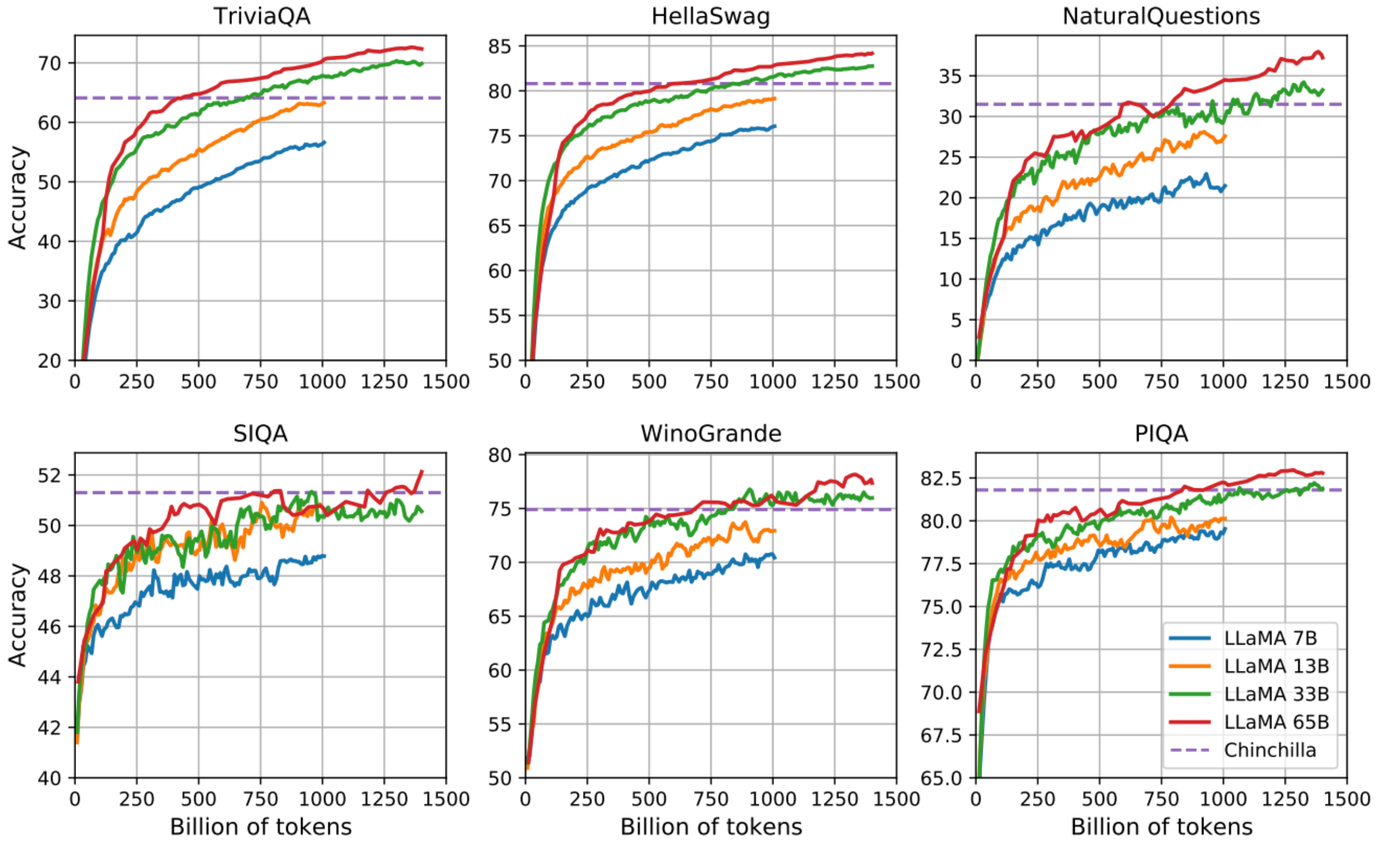
trained on 1~1.4T tokens



Touvron et al. (2023)

Llama (Preview)

- Smaller models continue to improve when trained longer on substantially more tokens.



Touvron et al. (2023)

Baidu Scaling Laws

- Scaling behavior wasn't unique to Transformers or LLMs.
- Irreducible error:

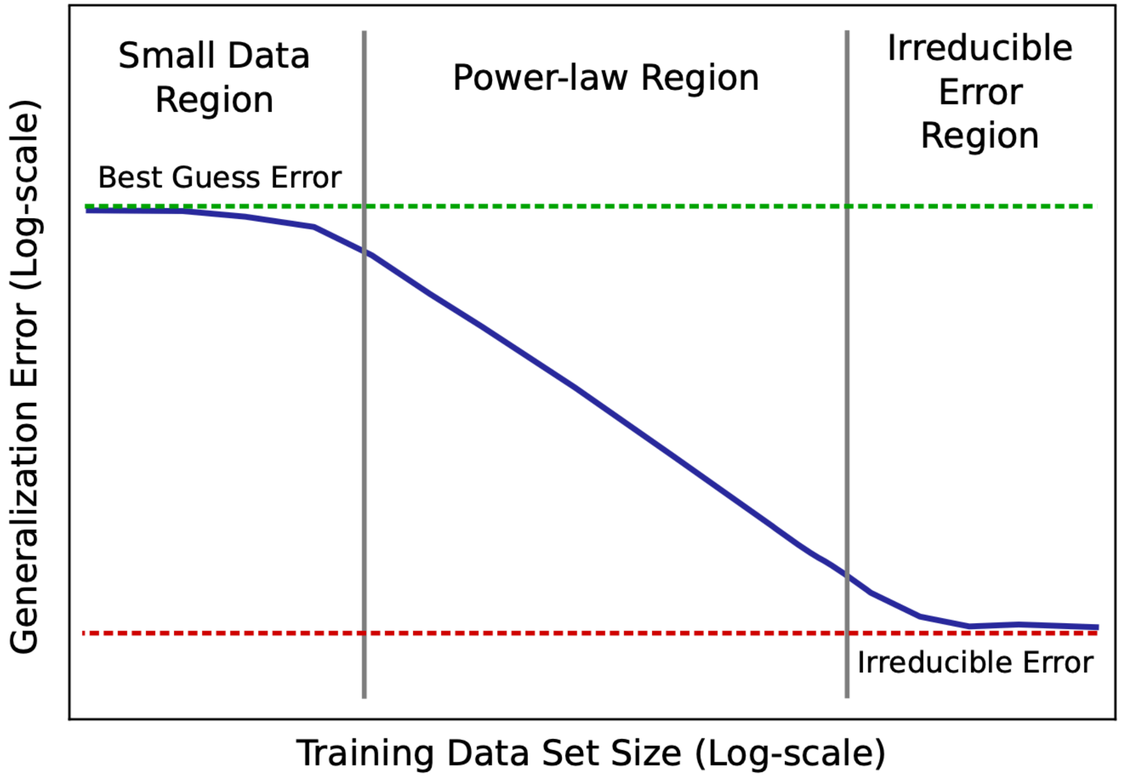
$$L(D) = L_\infty + AD^{-\alpha_D}$$

- e.g., cross-entropy loss in LM:

$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

true predicted

"I want a cup of ____"



Hestness et al. (2017)

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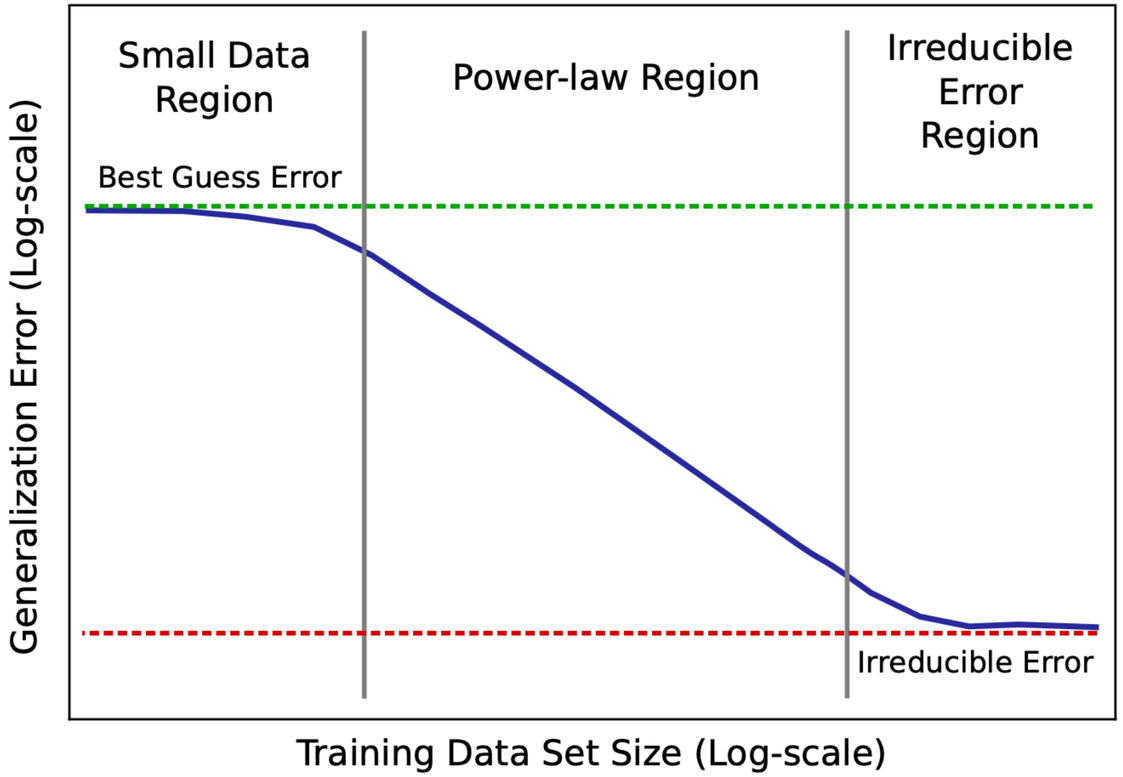
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Hestness et al. (2017)

Summary: Scaling Laws

- Hestness+ 2017: Scaling behavior wasn't unique to Transformers or LLMs.
- Kaplan+ 2020 (GPT-3): Prioritize model size, make models bigger
- Hoffmann+ 2022 (Chinchilla): Increase training data & model size equally
- Touvron+ 2023 (Llama-1): Smaller models continue to improve when trained on substantially more data, benefiting inference efficiency.

- Later: data-constrained scaling, sparse/MoE, test-time scaling, …
- And more: synthetic data, code, multimodal data, data filtering/curation, repeated data, post-training, reinforcement learning